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Review

A review on the applications of artificial neural network techniques for accelerating finite element analysis in the civil engineering domain

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ABSTRACT

Finite element (FE) modelling is widely recognised as the most powerful and foremost computational technique for analysing complex structural systems due to its highly efficient modelling and simulation capabilities. Despite the strengths, its computational demands restrict its ability of performing instantaneous computations and present substantial challenges for achieving real-time analyses results. However, the integration of artificial neural networks (ANNs) with FE modelling offers a simplified calculation process whilst accelerating the computational time substantially. In this light, current study conducts a worth and timely investigation on application of ANNs as a surrogate for real-time FE modelling in the civil engineering domain. ANNs can be integrated into the computational process of FE modelling in numerous ways and selecting the type of the neural network architecture for surrogate modelling and integrating it to represent the FE model are crucial. Even though that depends on the complexity of the model that is to be analysed, the graph neural networks (GNNs) showed an excellent performance in both analysing and visualising mesh-based FE modelling as it incorporates the underlying relationship among different elements in a FE mesh. ANNs have the potential to significantly accelerate forward calculation process and also, provide a more efficient approach to inverse analysis. However, such applications are seldom in the existing body of knowledge. It has been identified that predicting stress or displacements was the most common primary goal of most studies with a particular emphasis on addressing static structural behaviour.

1. Introduction

It is often required to analyze complex systems while dealing with designing, optimizing, and maintaining tasks of different structural elements in numerous engineering disciplines such as civil, automotive, chemical, biomedical, mechanical and many others. Various numerical analysis methods such as Finite Element method (FEM), Computational Fluid Dynamics (CFD), Boundary element method (BEM), Finite Difference method (FDM), Finite Volume method (FVM) etc. have been identified as ideal tools for such applications where the analytical solutions are more difficult or impossible to obtain. Amongst, the Finite element method (FEM) can be identified as the most popular and powerful computational technique that is used for structural analysis through highly efficient modelling and simulations whilst it is also a more common computational technique in electromagnetics, fluid and heat flows [107]. Besides, it acts effectively when solving problems with complex geometries, non-linear materials, multiple loading conditions, large deformations, and arbitrary boundary conditions [42]. The FEM discretizes a specific body into a mesh of small units known as finite elements (See Fig. 1) to derive equations for each finite element based on the underlying physics and then combines those equations with respect to the relationships between the finite elements to obtain a solution, rather than considering the entire body at once [100]. Accordingly, FE approximations converge through thousands of iterations and hence, the fidelity of FEM can be expected at a higher level.

FEM comprises a sequence of computations including three major steps [167]: I. Pre-processing – the FE mesh is created including material properties and boundary conditions, II. Solving – the solution is obtained based on desired loading patterns, III – Post-processing – further analyses on solutions are conducted in order to monitor the behavior of the structure in detail, See Fig. 2. Due to the complexity of the simulation process, different software packages such as ANSYS, Abaqus, Midas,

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COMSOL etc. have been developed and they are embedded with automatic mesh generation and scientific visualisation techniques that may assist in conducting complex modelling and interpretation of results.

However, the conventional approach of using computerized FE method includes certain drawbacks. The time and the computational power that is required for a Finite Element analysis strongly depends on the finite element mesh of the structure [100]. Whenever a higher number of degrees of freedom is available in a FE model, the required computational time and power of that model will also be increased which is, in fact an inherent characteristic of FE modelling [100]. Also, the required computational power and time may be doubled if it requires a redo of the analysis when the inputs get a slight change. Consequently, conducting a real-time simulation of a particular structure would be challenging due to these shortcomings. Accordingly, the investigations on surrogate methods for FEM, which can conduct real-time simulations consuming less computational power, while maintaining an accepted accuracy, are high in demand. Surrogate models are working with the concept of predicting the effectiveness of design goals whilst avoiding the direct calculations of results [132]. Applications of high-performance computers, parallel and pooling computing methods will obviously enhance the efficiency of simulations. However, that will not overcome the problems in sole means and a huge consideration should also be paid to other aspects which can reduce the complexity of the model while keeping the same accuracy. Several existing potential methods that can be used for such scenarios are Mass-Spring Systems [60], Model Order Reduction Techniques [11], Adaptive mesh refinement [140], and FEM-based training of Neural-Networks [54].

Springs connect concentrated masses together and the force will be transferred along the length direction in Mass-spring systems while transforming the structure into a truss and reducing its degrees of freedom (DoF) significantly. Hrennikoff [60] introduced this method in 1941 and later it was more prominently identified as a real-time simulation technique in surgery simulations and biomechanics [77,20]. However, they perform well only for the systems with high membrane stiffness and a negligible bending stiffness [168,166,119]. Also, it is challenging to define the topology of point masses to simulate the actual behavior of the physical object accurately [107]. Although the model order reduction technique (MOR) is far more popular to use for enhancing the accuracy of Multi-Body Systems (MBS), it can be used for real-time simulations as it simplifies a model [107]. MOR is more generally used to reduce the dimensions of Flexible Multi-Body systems (FMBS) by means of reducing coordinates of its deformations [102]and also by using modal vectors [7]. However, the mode shapes are required to be orthogonalized prior to the simulations and these approaches are only suitable for linearly behaved structures [107]. Hence, different techniques such as geometric stiffness matrix [29,106,108], modal warping [22]and restricting the non-linear behavior to one part of a structure [169]were applied to address the non-linearity of structures in reduced order modelling. Adaptive meshing densifies the FE mesh at the areas of interest and that concentrated mesh areas indicate regions with large gradients where more details of the mesh are required to be outlined [140]. Consequently, the computational effort will be reduced without sacrificing the accuracy of results while reducing the computational time substantially [27,96]. Hence, it will be beneficial to incorporate adaptive meshing in order to accommodate real-time FE modelling. Further, almost all the aforementioned methods directly reduce the degrees of freedom of the FE model to achieve fast computations. Hence, it could be noted as the most straightforward method of obtaining real-time conditions.

In the meanwhile, FEM-based traning of Neural Networks follows an indirect approach. In fact, the time-consuming complex computations are carried out offline under this method and those generated data will be used to train a neural network. Consequently, trained networks will respond to real-time inputs and generate outputs instantly. The use of ANNs is a sound method to map input data to output data with reference to a previously obtained dataset [54]. It learns from experiences and responds to different situations accordingly even though some experiences are incomplete and errorneous [134]. Unlike the other methods, any nonlinearity in the structure could be easily addressed in this method as those factors are concerned during the training phase. Initially, the FE analysis can be done with the use of a commercially available software and a data set will be generated including the desired inputs and outputs. Subsequently, an artificial neural network (ANN) will be trained and validated using that data set. Ultimately, the model can be setup for real-time simulations. The behaviour of the ANN totally depends on the training data set and it is utterly important to consider about its validity domain prior to its application [127]. Once, trained, it performs fast and highly accurate computations [23]. Further, a conventional neural network deals with numbers and hence, it does not showcase the underlying theory of a simulation. Later a hybrid method introduced which performs the training of a neural network utilising a FEM-based custom loss function and verified the accuracy of that method using three case studies on forward and inverse problems which have been conducted on simple 2D structures while stating the requirement of a validation of this method for more complex structures [111]. Similarly, vast number of applications present the possibility of utlising neural networks for fast and accurate finite element simulations [72,155,117,159]. In this light, ANNs can be identified as extremely useful concept for fast computations of FE modelling and it is clear that they will be indeed important for the future phase of Finite element modelling. Furthermore, ANNs fall under the category of supervised learning within machine learning (ML) algorithms. In addition to ANNs, other machine learning algorithms such as support vecor machines [39], ensemble models [133], gradient boosting [18]and Bayesian probabilistic models [92,93,173] have also been explored for integration with FEM. However, their applications are often tailored to specific use cases, with limitations in addressing the broader complexity of FEM integration. These methods, though robust in their respective domains, lack the

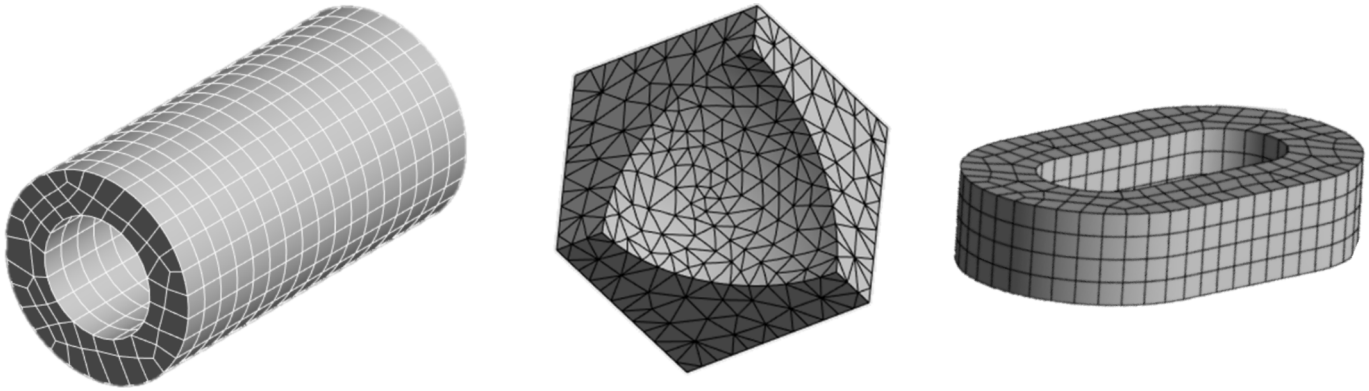


Fig. 1. Finite Element mesh of various objects.

adaptability and scalability of neural networks when applied to high dimensional FEM problems. In contrast, ANNs, offer unparalleled flexibility in addressing complex geometries and non-linear relationships, as demonstrated in this review.

Both the concepts of FE modelling and neural networks have demonstrated remarkable individual performances in predictive tasks. On one hand, FE modelling has been a fundamental tool in civil engineering for simulating and predicting structural behaviour under various loading and boundary conditions with high accuracy. However, its computational demands often restricts its use in real-time applications, particularly for large-scale or dynamic systems. On the other hand, ANNs have emerged as highly efficient tools in data science, capable of delivering near-instantaneous predictions once trained, albeit with some limitations in capturing the full complexity of physical systems. By combining the strengths of Fem's accuracy and ANNs' efficiency, this integration holds the potential to overcome computational challenges and enable more practical real-time applications in structural analysis. However, the potential synergy between these methods when combined remains largely unexplored, and their collective performance is not yet fully understood. Neural networks encompass a range of types, including the Multi-layer Perceptron (MLP), Recurrent Neural Networks (RNN), Convolutional Neural Networks (CNN), Graph Neural Networks (GNN), and Physics-Informed Neural Networks (PINN). Each of these architectures offers distinct approaches to integrating with finite element methods (FEM), addressing different computational goals. Numerous studies have explored the promising potential of incorporating neural networks into FE modeling. However, it remains unclear which neural network architecture performs optimally for specific tasks, and the most effective strategies for combining these two methods to enhance computational efficiency are yet to be fully understood[84,83]. Also, the integration of ANNs with FE modeling in the civil engineering domain lacks a universal framework, resulting in fragmented methodologies and inconsistent outcomes.Moreover, the critical factors that should be considered when developing a neural network as a surrogate for FE models are not well established. The literature rarely provides a comprehensive analysis of the significant benefits and drawbacks of each approach in surrogate FE modeling. This lack of clarity presents a challenge for researchers seeking to optimize the combined performance of neural networks and FEM.

In this context, this paper investigates the various approaches to ANN integration, evaluates their effectiveness, and explores the conditions under which they are most applicable. This includes a comparison of the five most popular neural networks in surrogating FE modelling. The study highlights the advantages and limitations of each method, offering a comparative perspective on their application in the context of FEM. Furthermore, this research paves the way for future studies by identifying key areas where parameter optimization could further enhance the computational efficiency of neural network-FEM integration. Consequently, this review offers valuable insights into the development of more efficient surrogate models and contributes to the growing body of knowledge in this evolving field. As the key contributions of this study it includes:

- Reviews on methodologies used for integrating ANNs with FE models in the civil engineering domain.
- Provides a comparative analysis of alternatice machine learning approaches, including support vector machines and ensemble models
- Discusses the practical challenges and limitations of ANN-based surrogates, such as data requirements, generalization issues, and computational trade-offs.
- Identifies gaps in the existing research and suggests future directions, including hybrid modeling techniques and the use of physics-informed neural networks (PINNs).

2. Research methodology

This review was undertaken with the primary objective of promoting attention on the application of neural network models to surrogate FE modelling in the civil engineering domain, particularly in relation to various mechanical systems governed by solid mechanics. This paper presents an insight review on different methods which can be used to optimize the efficacy of ANNs to perform real-time FE modelling. On this basis, the current study initially examined around 200 journal articles, conference papers and book chapters which were written on the use of ANNs for FE modelling for the purpose of accelerating computatonal time, and irrelevant articles were excluded considering the abstract, introduction and conclusion of those research articles. Different combinations of keywords such as "real-time finite element modelling", "surrogate modelling" and "neural networks" were used to explore the potential literature in search engines including Google Scholar, IEEE Xplore, ScienceDirect, ProQuest, JSTOR and Scopus. The research methodology has been presented in Fig. 3.

Upon selection of the peer-reviwed journals and conference papers, the number of publications were analysed per year, see Fig. 4. The research studies that were published in between 2008 and 2024 were mainly reviewed in the current study except few studies that were publised based on theories even before the mentioned period. It is noticeable that the number of publications has been significantly increased in recent years when compared to earlier years showing the growing interest and demand on this field.

Further, based on the developed ANN models in the literature, it was decided to explore the possible types of ANNs for the real-time FE modelling together with their current applications. Consequently, section 3 outlines the significance of the research whilst section 4 explains the concept of ANNs, section 5 presents the ways of integrating FEM and ANNs and section 6 explains the applications of them in FE modelling in the civil engineering domain. Ultimately, section 7 summarizes the ANN model implementation strategies whilst section 8 includes the findings and presents the possible research gaps in this scenario highlighting the possible future work in this study area.

3. Research significance

The Finite element method can be identified as the most powerful

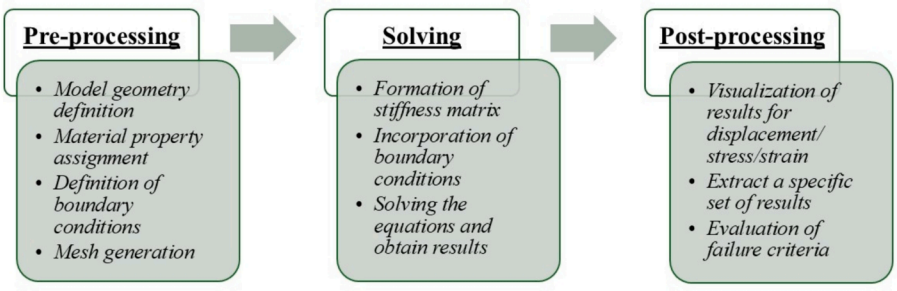


Fig. 2. Steps of FE analysis process.

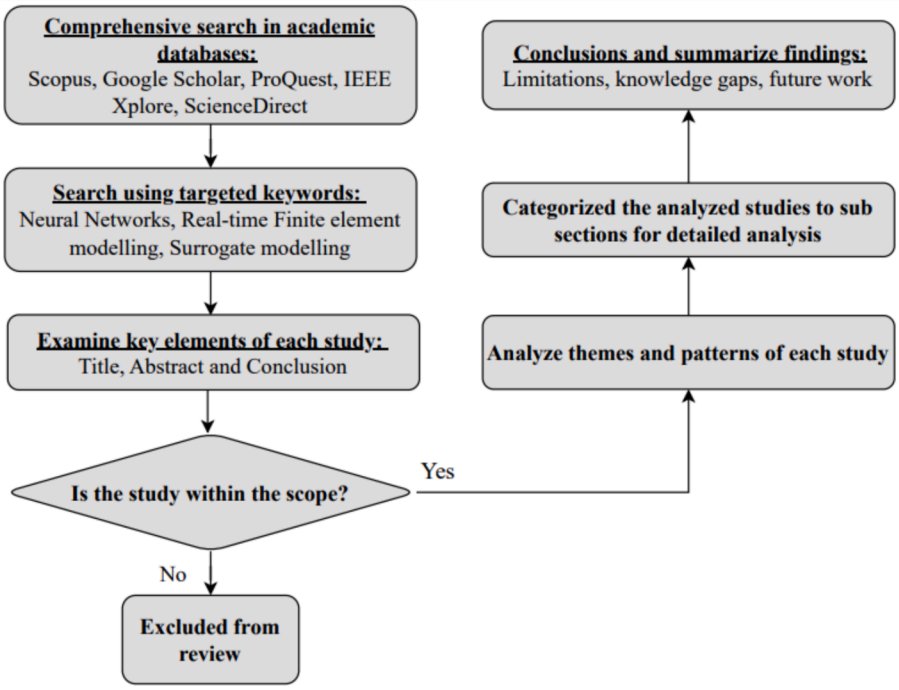


Fig. 3. Review Methodology.



Fig. 4. Number of publications from 2008 to 2023.

and the most popular computational technique across various fields, particularly in structural engineering. Despite its accuracy and versatility, FEM-based analyses are computationally intensive, requiring significant time and resources to achieve high-fidelity results. This limitation poses challenges for real-time applications, such as structural health monitoring (SHM) and digital twins, where immediate assessments of structural responses are critical. For instance, monitoring the real-time behaviour of a bridge under continuous and variable loads, environmental challenges, and extreme conditions is vital for preventing sudden failures and prioritizing rehabilitation efforts. However, the high computational cost of real-time FEM simulations makes them impractical for such applications. To address this challenge, surrogate models have emerged as a promising solution for reducing computational complexity while maintaining the accuracy of FEM. Among these approaches, artificial neural networks (ANNs) have gained significant attention due to their ability to learn complex patterns from extensive datasets and provide near-instantaneous predictions once trained. These characteristics make ANNs a powerful tool for simulating real-time structural behaviour, enabling intelligent maintenance decisions and enhancing the efficiency of SHM and digital twin implementations.

Despite the growing interest in ANN-based surrogates, existing reviews in this domain often lack a comprehensive analysis of the integration of neural networks with FEM. Additionally, the potential of emerging machine learning (ML) technologies, such as graph neural networks (GNNs) and physics-informed neural networks (PINNs), remain underexplored. A critical synthesis of current methodologies and a detailed evaluation of their strengths and limitations are necessary to identify best practices and guide future advancements in this field. This review addresses these gaps by providing a thorough examination of the state-of-the-art ANN methodologies used as surrogates for FEM, with a

specific focus on their application to real-time structural health monitoring and stress predictions in civil infrastructure systems. The contributions of this review are threefold:

- Presents a critical synthesis of existing approaches for ANN-based FEM surrogates, highlighting the challenges and opportunities associated with their implementation.
- Identifies best practices for selecting and implementing ANN architectures, such as multi-layer perceptron (MLPs), recurrent neural network (RNNs), convolutional neural networks (CNNs), and graph neural networks (GNNs), for specific structural applications.
- Proposes future research directions, including the development of domain-specific datasets, incorporation of uncertainty quantification techniques, and leveraging advanced hardware technologies to enhance training efficiency and scalability.

By addressing these aspects, this review provides a valuable resource for researchers and practitioners seeking to advance the application of neural networks in FEM-based simulations. The findings of this study will pave the way for more efficient real-time structural analysis, intelligent infrastructure monitoring, and the broader adoption of digital twin technologies in civil engineering.

4. Artificial neural networks

Artificial Intelligence (AI) is a broad field that encompasses the development of intelligent systems capable of performing tasks that typically require human intelligence. While Machine Learning (ML) is a subset of AI that focuses on algorithms and statistical models that enable computers to learn and make predictions or decisions without being explicitly programmed (Vita et al. 2023). However, Deep Learning (DL) is a specialized branch of ML that utilizes Artificial Neural Networks (ANN), particularly Deep Neural Networks (DNN) with multiple layers, to learn and extract complex patterns or features from data (Rengasamy et al. 2020) as the same way as the human brain acts. The Nobel Prize-winning research of Hubel and Wiesel on the cat's primary visual cortex served as the foundation for neural networks[63]. It has been found that the hierarchical structure of the neural networks is consisted of layers of cells in the visual cortex and also, higher order cells are more prone to complex characteristics of the stimulus patterns whilst possessing a strong receptive filed and being less sensitive to deviations of stimulus patterns. Consequently, the pattern recognition ability of human brains was more emphasized and that finding was adopted for several subsequent studies [137,71,44,36]. However, the majority of those models were significantly sensitive to slight changes in input patterns and were unable to mimic the pattern recognition ability of human brains accurately. As a respond for that drawback, the first mathematical model of neural networks which is known as the "neocognitron" was introduced with the concept of unsupervised learning [37]which was in fact a revolutionary change in the field of investigations on neural networks. Subsequently, the neural networks were evolved for the past few decades and it has recently become as the most powerful technique which is extremely useful for pattern recognition and prediction tasks in the field of data science engineering [16].

A system of artificial neurons develops a neural network whilst "nodes" play the most vital role of acting as the basic processing unit of the whole network [1,52]. The processing capacity of a neural network solely depends on the connections in between nodes and a weight is given for each and every interconnection to reflect their contribution to the whole network. Weights are determined through a series of training trails and ultimately a neural network will be able to produce a set of outputs which will be resulted by mapping to a given set of inputs [25]. The basis for the machine learning is the optimization and similarly, neural networks also utilize a compositional function $\underset{A_j}{\operatorname{argmin}}\,(f_M(A_M, \cdots \cdots, \mathrm{f}_2(\mathrm{A}_2, \mathrm{f}_1(\mathrm{A}_1, \mathrm{x}))\cdots) + \lambda g(\mathrm{A}_j))$ for its optimization. In here, A_k refers to the weights of the interconnections whilst $\lambda g(\mathrm{A}_j)$ denotes the regularization [16]. Consequently, an ANN is based on a gradient based optimization algorithm and the optimization process will be done in order to minimize the difference in between the output of the network and the target output which is in fact obtained via minimizing a scalar loss function and the process is also referred to as the training of the neural network [25]. In the meanwhile, the back propagation [139]and the stochastic gradient descent algorithms [16]can be identified as the most prominent algorithms in estimating gradients of neural networks.

The architecture of ANNs is always determined based on the data

Table 1
Applications of different neural network architectures.

Type of the Neural Network	Characteristics	Specific Applications
Multi-layer Perceptron (MLP)	• Information flows from input nodes to output nodes. • Layered architecture – an input layer, few hidden layers and output layer	• Pattern Recognition • Classification • Regression • Time series analysis
Radial Basis Function (RBF)	• Similar to MLP but uses a radial basis activation function • Parameters are optimized using methods likes clustering or least squares	• System Modelling • Time series prediction • Function approximation
Convolutional Neural Network (CNN)	• Processing grid-like structures such as images or time series	• Classification • Crack detection • Damage detection
Graph Neural Network (GNN)	• Dataset will be analyzed corresponding to a graph structure	• Network vulnerability assessment • Recommendation systems • Stress predictions of FE mesh models
Recurrent Neural Network (RNN)	• Handling sequential data • Feedback connections • Maintain an internal memory	• Long term prediction of structural conditions • Time series analysis
Long/Short Term Memory (LSTM)	• A type of RNN • Contains memory cells and gating mechanisms	• Time series analysis • Rainfall-runoff modelling • Prediction of structural conditions
Autoencoder (AE)	• Consists of an encoder and decoder • Compress data by encoding	• Dimensionality reduction • Anomaly detection • Image Denoising
Physics Informed Neural Network (PINN)	• Training includes physics-based equations or constraints	• Solving partial differential equations in fluid dynamics, structural mechanics and heat transfer
Attention Network	• Visualizing corresponding input features to output features • Focuses on specific part of input data using attention mechanisms	• Non-destructive testing analysis • Structural Damage localization • Estimation of remaining useful life

types and the intended output of the network. As new variations and combinations are continuously developing, it is immensely challenging to estimate the number of available types of neural networks. It is possible to customize the architecture of a neural network according to the purpose. However, some of the identified types of NNs along with their main characteristics, specific applications and several examples in civil engineering domain have been showed in Table 1. Among these, the MLP (Multi- Layer Perceptron), RBF (Radial- Basis Function), Physics-informed Neural Networks (PINNs), Convolutional Neural Networks (CNNs), Graph Neural Networks (GNNs), Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are commonly used as surrogates for FE modelling in civil engineering. RBF networks differ from architectures like MLPs as they often use clustering or least square methods for parameter optimization rather than gradient descent or backpropagation[17] while PINNs can also be identified in different ANN architectures with embedded physics-based principles usually in terms of a loss function [111]. Hence, those neural networks will be explained in detail under section 05. CNNs excel in image and pattern recognition tasks, while RNNs and LSTMs are well-suited for sequence-based data analysis, such as time series or text processing. ANN models, including CNNs, RNNs, and LSTM networks, have shown promising applications in big data prediction. However, these methods require a significant volume of labelled data samples, which may not be available in situations where preventive replacements are common (Dangut et al., 2022). This process involves the use of an Autoencoder (AE) to learn the normal system behaviour from the unlabeled data samples and to detect deviations resulting from increasing degradation. However, real-world applications often involve different entities operating under varied conditions, leading to different data distributions and characteristics, which presents a significant challenge in generalizing learned knowledge from training data to testing data.

5. Integration of ANNs with FEM

5.1. Approaches for integrating ANNs with FEM

Different supervised machine learning algorithms are commonly being used as surrogates for FE analysis. Amongst, the Artificial Neural Networks demonstrate superior performance. For instance, Badarinath et al. [3] analyzed an one-dimensional beam by comparing its analysis results on three different machine learning models such as extreme gradient boosting trees, Random forests and Artificial Neural Networks (ANNs) and found that ANN was the most predictive out of the tested models. Vardhan and Sztipanovits [161] also confirmed the efficiency of ANNs by comparing its performance with Random Forests and gradient boost regression models in designing sub-sea pressure vessels using FEM. Similarly, the efficiency of ANNs has been verified by comparing with other ML algorithms elsewhere [68]. The use of neural networks in finite element analysis has garnered significant interest, with numerous studies demonstrating their effectiveness in reducing computational time while analyzing models. FEM and ANN are both powerful prediction tools in different domains and the integration of NNs with FEMs can be achieved through various approaches, each offering unique benefits and challenges. Table 2 summarizes the key integration methods, their characteristics, and some of the major references from which the information was derived.

As mentioned in Table 2, the direct integration method involves either replacing FEM simulations with NN based surrogates or embedding NNs directly within the FEM framework. This approach offers significant computational time savings and cost reductions, making it highly effective for applications requiring quick approximations. However, maintaining physical consistency and ensuring fidelity to physical laws remain substantial challenges. Section 5.2 explores various methods for directly integrating NNs with FEMs, while section 6.1 delves deeper into how this integration can be achieved using different types of ANNs. The hybrid models leverage FEM-based loss functions in ANN training to preserve physical interpretability while benefitting from the flexibility of data-driven models. These methods strike a balance between computational efficiency and adherence to physical principles but come with challenges such as complex implementation and high computational requirements during training. Section 6.2 explores more about hybrid models in terms of PINNs for surrogating FEMs. The pre/post-processing integration method utilizes ANNs to enhance FEM workflows, particularly in tasks like adaptive mesh refinement and preprocessing of input data. This approach significantly improves the accuracy and efficiency of FEM analyses by focusing computational resources on critical areas. However, its effectiveness depends heavily on the availability of accurate and representative training datasets, which can be difficult to obtain in some cases. Overall, these integration methods demonstrate the diverse ways in which NNs complement and enhance FEM, each with its unique strengths and limitations while highlighting the potential for combining the accuracy of FEM with the efficiency and adaptability of ANNs in engineering applications. When comparing these integration methods, direct integration and hybrid modeling approaches, such as PINNs, demonstrate significant benefits and represent timely and valuable areas for investigation. Consequently, this paper primarily focuses on the direct integration of ANNs with FEMs to evaluate their impact on accelerating computational times.

5.2. Direct integration of ANNs with FEM

The direct integration of ANNs with FEM can be obtained under three main distinct approaches. The first approach is based on embedding the ANN directly into the FEM architecture. FEM consists of series of complex analysis processes and each of them contribute for the time consumption of the overall process. Thus, some of those time-consuming steps can be simulated using trained ANN models and they can be directly fed into the main FEM framework in order to reduce the computational time substantially. For instance, trained ANN models can be used to simulate complex material models [66,151]and it has been identified that ANN embedded material models is even capable of reducing the processing time of FEM significantly whilst keeping the prediction offset less than 5 % [50]. Further, as another example, ANNs have identified to be directly integrated into the FEM workflow to predict stochastic responses, such as vibration and buckling, of functionally graded (FGP) microplates, enhancing the ability of the model to address complex problems like stochastic response prediction and optimization [158]. Similarly, the second approach develops an empirical equation based on a trained ANN model using data generated from FEM.

Table 2
Different Integration methods in between ANN and FEM.

Reference	Integration method	Description	Advantages	Challenges
[47,79,67]	Direct integration	NNs replace FEM for approximate simulations or embedded to the FE model directly	Reduces computational time while drastically reducing computational cost	Maintaining physical consistency, loss of physical fidelity in approximation
[160,21,162]	Hybrid methods	Combines FEM-based loss functions with NNs	Retains physical interpretability	Complex implementation and training
[6,129,138]	Pre/post-processing	NNs for preprocessing input data or adaptive mesh refinement	Improves FEM efficiency and accuracy	Requires accurate training datasets

Following to the training of an ANN model, the optimum values of the weights and biases can be obtained to define an empirical equation. Thus, the required input values can be substituted to the equations to obtain the resultant values easily [152]. The third approach is the most common and popular approach as it utilizes the FEM to produce training data to build an ANN model in between the desired inputs and outputs. Once trained, the required input values will be given to the trained model to generate the outputs. Thereby, the analyses will be completed instantaneously. Hence, such an approach is highly recommended for real-time analyses and more popular amongst parametric studies [101,48,144]. However, as this kind of analysis deals with numbers, it does not showcase the underlying theory of a simulation. Hence, some researchers have focussed on developing hybrid FEM-NN models that are based on the FEM-based custom loss functions to incorporate the underlying theory of a physical system in terms of partial differential equations (PDEs) [115,111]. Either way, the ANNs highly rely upon a training dataset and hence, a significant consideration should be paid on the validity domain of the dataset prior to its application [127]. The benefits and drawbacks of each integration method are shown in Fig. 5.

Also, the third approach of ANNs is commonly being used as a surrogate for real-time FE modelling when solving inverse problems where the structural response is measured instead of the loads applying on the structure [51] and such an approach is exceptionally important in the facet of SHM of civil infrastructure systems. Liu [98] presented a comparative study and emphasized the ability of applying neural networks for both forward and inverse problems. They corresponded the shapes of deep nets of them to a trumpet shape where the forward problems produce large number of outputs whilst the inverse problems require large number of inputs. A rectangular plate was modelled under the case study to predict the displacements and to predict the material properties respectively. The maximum error of the inverse model was about 0.84 % and it was able to predict the desired outputs within 0.0001 s emphasizing both the high accuracy and real-time performance of trained ANNs.

5.3. Pre/post-processing (Mesh refinement)

Furthermore, finite element (FE) modelling relies on an FE mesh that mathematically represents the entire model. Mesh generation is a crucial step in FE analysis, as the quality of the mesh significantly impacts the accuracy and efficiency of the computations [100]. Beyond the direct integration of ANNs for computational analysis in FEM, their application has also been explored in the context of mesh generation. Adaptive meshing, in particular, is recognized as an effective method to reduce the computational intensity of FE modelling [140]. However, balancing the computational cost and accuracy remains a significant challenge. To address this, Jilani et al. [70] proposed a novel method integrating self-organizing maps (SOM), a type of ANNs for adaptive mesh generation. Their approach uses SOMs to generate an initial mesh by mapping a computational domain onto a physical domain, adjusting mesh nodes based on density functions. FE and adaptivity computations then guide SOMs to refine the mesh, concentrating smaller elements in critical areas and coarser elements elsewhere, enhancing efficiency. This approach is particularly suited to two-dimensional triangular elements and offers potential for extension to other geometries and hybrid optimization methods. Also, its focus on convex domains, computational overhead from SOM processes, and reliance on basic neighborhood kernels are few other identified limitations. Similarly, in another study Bahreininejad et al. [4] applied neural networks to optimize mesh partitioning using recursive bisection, ensuring efficient subdomain decomposition for parallel computations whilst Topping et al. [153] also demonstrated the use of ANNs to predict the number of elements for adaptive mesh refinement, assisting with load balancing and parallel FE analysis, and Topping et al. [154] combined neural networks with genetic algorithms for efficient mesh partitioning in parallel computations. Both approaches underscore the potential of ANNs to streamline FE workflows, although challenges such as parameter sensitivity and computational overhead remain. Also, another framework, referred to as FAN-ROM which combines adaptive mesh refinement (AMR) with ANNs to enhance the accuracy and efficiency of FEM, introduced a novel reduced-order modelling approach for FE mesh preparation [6]. This method significantly improved the accuracy of coarse mesh models and allowing them to approximate the behavior of high-fidelity models with minimal error while providing an efficient alternative for FEM analysis.

Recently, the focus has shifted towards leveraging Graph Neural Networks (GNNs) for adaptive mesh refinement (AMR), owing to their

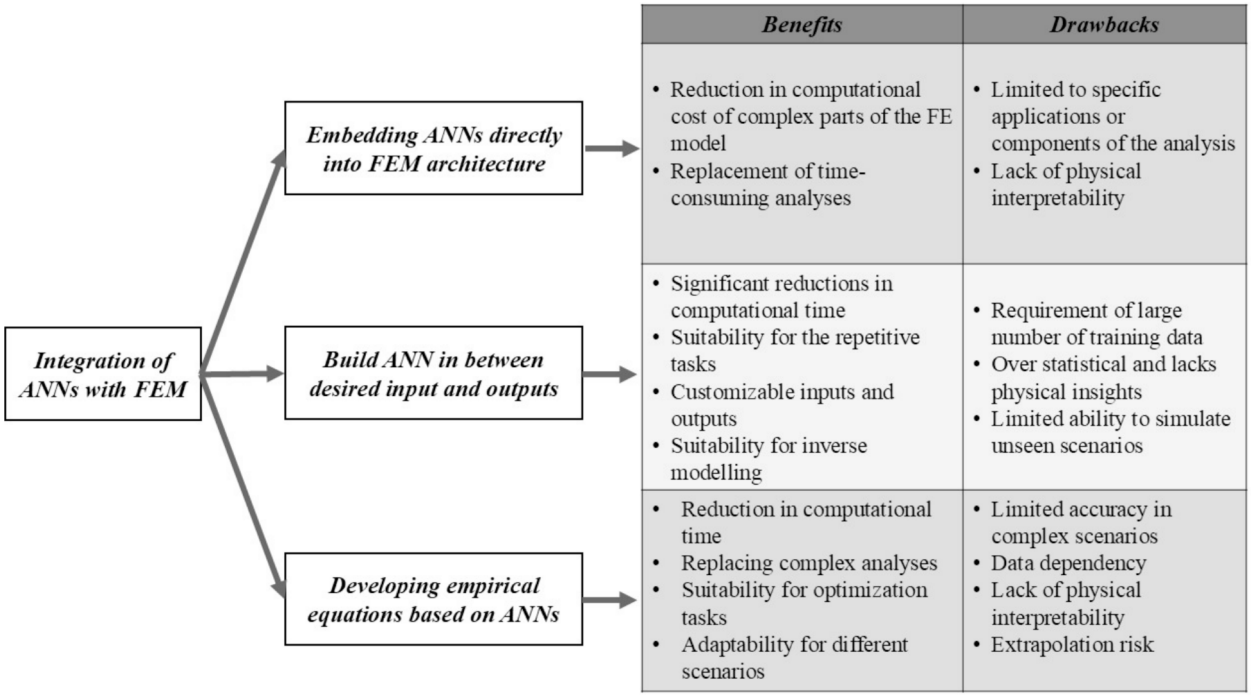


Fig. 5. Benefits and drawbacks of different integration methods.

inherent ability to handle graph-structured data, which naturally aligns with the mesh-based nature of finite element modeling (FEM) [109]. Unlike traditional machine learning approaches, GNNs can directly operate on unstructured mesh data, enabling more efficient and accurate refinement processes [146]. For instance, in Pelissier et al. [129], a GNN-based framework called Adaptnet was introduced, which utilizes two interconnected GNNs, Meshnet and Graphnet, to predict initial mesh parameters and anisotropic adaptation metrics, respectively. This approach enhances computational efficiency by refining mesh resolution in critical regions without requiring manual intervention, significantly improving accuracy in structural and fluid mechanics problems. Rowbottom et al. [138] presented a method termed G-adaptivity, which combines graph neural diffusion architectures with a differentiable FEM solver (difFEM). This technique optimizes mesh point locations directly to minimize FEM solution errors, achieving superior accuracy and efficiency compared to traditional AMR methods, particularly for r-adaptive meshing scenarios. These advancements highlight the growing potential of GNNs to revolutionize adaptive mesh refinement by directly integrating data-driven insights into FEM workflows, paving the way for more efficient and scalable solutions.

Further, the integration of ANNs with FEM can extended to applications such as structural health monitoring (SHM) and damage detection [112,121,156,122]. These applications highlight the versatility of ANN-FEM combinations in addressing complex engineering challenges [176,178]However, the primary focus of this review is on studies where ANNs have been specifically employed as a surrogate for FEM and analyzing structures, with the objective of reducing the computational intensity associated with structural analyses in the civil engineering domain.

6. ANN architectures and frameworks for FEM applications

Various ANN architectures and frameworks have been employed to integrate neural networks with FEM, leveraging their shared principles for computational efficiency [148]. This section provides a comprehensive overview of some of the most commonly used models, including their suitability for fast and accurate FEM computations. The discussion focuses on analyzing the applicability of each type for real-time FE analysis, with the goal of supporting real-time structural integrity assessment of infrastructure systems.

6.1. Application of neural network architectures for FEM surrogates

The direct application of neural network architectures as FEM surrogates are explained in this section considering MLPs, RNNs, CNNs and GNNs.

6.1.1. MLP (Multi- layer Perceptron)

Multi-layer Perceptron (MLP) based neural networks can be identified as the most popular neural network and are widely used in multiple tasks such as regression, classification, and pattern recognition tasks due to their versatility. It can be identified as a simple fully connected type of feedforward ANN and is typically composed of three layers: an input layer, a hidden layer and an output layer, see Fig. 6., x1, x2.. $\mathbf{x_n}$ are inputs and **y** is the output.

However, the number of layers can be increased by introducing different numbers of hidden layers to the network. Also, in MLPs the input layer matches the input variables, while the output layer aligns with the quantities of interest. However, to maintain the simplicity, the hidden layer typically has a minimal number of nodes since more nodes increase parameter complexity and data requirements for accurate estimation [32]. Also, Nodes or neurons in each layer, also known as processing units, connects with the neurons only in the next layer [47]. Due to the feedforward nature, MLPs do not have the history-dependent prediction capacities [85].

The integration of MLPs with FEM has been done for both parametric and non-parametric studies whilst it can also be noticed for both forward and inverse analyses problems. MLP regressor models are commonly being identified as surrogates for FEM and also, some researchers even have compared the efficacy of MLP regressors with some other machine learning algorithms such as random forest, gradient boosting, k-nearest neighbor, support vector and gaussian process regressor in surrogating FE analyses and found that the performance of MLPs is superior than other machine learning models [3,59]. MLPs let the capacity to define the inputs based on the intended purpose of the study and allow to build up a relationship between the inputs and desired outputs which has been obtained via FE analysis. Table 3 shows a series of applications of MLPs for FE analysis conducted in the civil engineering domain and they have reflected an exceptional performance (In Table 2, Lr – Learning rate, RMSE – Root mean square error, MSE – Mean square error, MAE – Mean absolute error). As per recent literature, MLPs have been used for both forward and inverse structural analysis even though a limited number of literatures are available for inverse problems [112]. However, it has

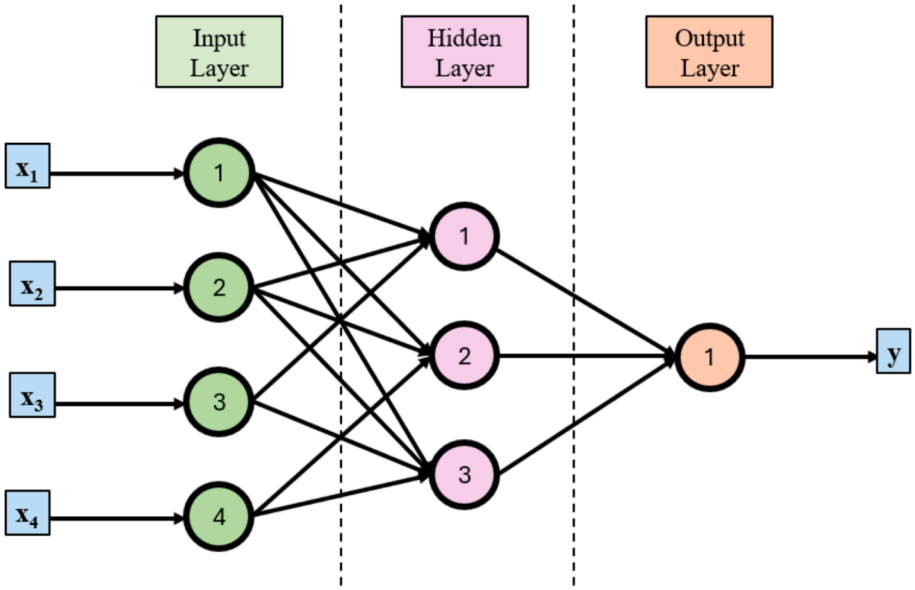


Fig. 6. Schematic of Multi-layer Perceptron Network.

Table 3
Applications of MLPs in surrogating FE modelling.

Author	Purpose	Size of the Data set	Data Preparation	Structure of the ML model	Activation Function	Model Training	Inference Time	Performance
Ferrario et al. [32]	To estimate the maximum structural top displacement of a masonry building	168 data records, 70 % for training, 15 % for validation and 15 % for testing	Non-linear dynamic FE analysis was done for a masonry building	3-layer model with one hidden layer	Not mentioned	Trained by the error back propagation algorithm	Less than a second	Test for various sizes of datasets. Minimum RMSE was 0.136
Conceição António and Rasheed [24]	To identify the elastic properties of materials using two case studies: the off-axis tensile test and hinged laminated composite plate	27 training datasets	Uniform Design Method used to determine a set of design points in a particular interval and FEM was used to determine the displacement field for each point	3-layer model with one hidden layer. Input layer – 4 neurons. Hidden layer – 5 neuronsOutput layer – 15 neurons	Sigmoid function on hidden layer and linear function on output layer	Using genetic algorithm and concluded after 30,000 generations of the genetic algorithm	Not mentioned	Maximum of 1.38 % and 3.29 % of relative error for the two case studies respectively
Kononenko and Kononenko [82]	To predict the maximum displacement in 3D stainless steel beam	1990 data records, training and testing data were selected based on random selection criterion	FE analysis of a 3D stainless steel beam under harmonically varying external loading	4-layer model with one input, 2 hidden layers with 200 neurons and the output layer with 3 neurons	Not mentioned	Adam optimization algorithm, extension to stochastic gradient descent implemented in TensorFlow was used to minimize the MSE	Not mentioned	Showed a really close relationship between ANN and FEM results using graphs without quantifying the error
Liu [98]	Idealized model problem of a composite laminated rectangular plate	2400 samples	2D plate FE modelling using a downward pressure on the top edge fully constraining	3-layer model with one hidden layer	Different activation functions	Trained until the error reduces to 1 %	Time required for prediction of 100 test samples	MSE
	- Forward Problem	2300 for training and 100 for testing	the left edge of the plate in ABAQUS	Input layer – 4 neurons Hidden layer – 15 neuronsOutput layer – 25 neurons	Linear function	Training time – 250 s	0.012 s	0.01
	- Inverse Problem			Input layer – 25 neurons Hidden layer – 15 neuronsOutput layer – 4 neurons	ReLU function	Training time – 486 min	0.01 s	0.0084
Zeroual et. al (2019)	To obtain the factor of safety (FOS) of small dams under extreme events	1372 data points, 70 % training, 15 % testing and 15 % validation	2D plane stress analysis of perfectly plastic soils considering Mohr-Coulomb failure criterion using GEO5 (2017)	3-layer model with one hidden layer Input layer – 11 neurons Hidden layer – 21 neuronsOutput layer – 4 Neurons	Tan-sigmoid transfer function	Levenberg–Marquardt (ML) was the optimization function, epochs = 1000, Lr = 0.001	Not calculated	MAE for four FOS, FOS1 − 0.01372876 FOS2 − 0.00315325 FOS3 − 0.00945855 FOS4 − 0.01270658
Badarinath et al. [3]	To predict the stress distribution of a one-dimensional aluminum beam	1.6 million datapoints, 80 % training and 20 % testing	One-dimensional, linear, flexible dynamic analysis of a beam considering a mesh of 40 Timoshenko beam elements	3-layer model with one hidden layer Input layer – 6 neurons Hidden layer – 1952 neuronsOutput layer – 1 neuron	ReLU Activation function	Trained for 50 epochs. Lr = 0.001, using RMSProp optimizer in Python	Showed the real-time performance	MAE of 2.87
Hoffer et al. [59]	To determine the stress, strain and deflection	0.1 validation split for each case	General static FEM simulation using Abaqus	Different number of hidden layers each with 100 neurons	ReLU Activation function	Epochs = 100,000, early stopping patience = 5000 for plate and beam and 7500 for block	Required time was	R2 scores
	- Plate	Training – 56 %, Testing – 44 %		3 hidden layers			0.0523 s	0.9900
	- Beam	Training – 56 %, Testing – 44 %		2 hidden layers			0.0781 s	0.6682
	- Block	Training – 62 %, Testing – 38 %		4 hidden layers			0.0781 s	0.5562
Le-Duc et al. [86]	For parametric simulation of static problems	Two datasets – clean and noisy	Geometric and parametric features were inputs and nodal displacements	4-layer model with two hidden layers with 100 nodes on each	ReLU activation function	Adam optimizer is used with maximum epoch of 500 for the plate and 2000 for the rest. Lr =	Not mentioned	MAE for each scenario
	- Planar 10 bar truss	9000 for training	considered via high fidelity					–
	- Spatial 72 bar truss	and 1000 for testing	FEM simulators			0.001, exponential decay rates = 0.9 and 0.999		1.02E-4

(*continued on next page*)

Table 3 (*continued*)

Author	Purpose	Size of the Data set	Data Preparation	Structure of the ML model	Activation Function	Model Training	Inference Time	Performance
	- Spring supported Euler-Bernoulli beam - Squared Reissner-mindlin plate							8.55E-2
Guan et al. [47]	To predict the constitutive behavior of soils	800 for training and 200 for testing 128 datasets with 0.1 validation split	2D plane stress analysis disregarding the gravity	5-layer model with three hidden layers Input layer – 6 neurons. Hidden layers – each with 128 neuronsOutput layer – 3 neurons	ReLU Activation function	Adam optimizer, epochs = 1000, Lr = 0.001	Significant acceleration was observed in the calculation time	1.90E-3 MAE of 0.6
Jayasinghe et al. [67]	To predict the nodal displacement	10,000 datasets, 80 % training, 20 % testing and 20 % of testing data – validation	3D Static analysis using ANSYS for different 10,000 loading patterns	4-layer model with one hidden layer Input layer – 7 neurons, Hidden layers – one with 64 neurons and other with 32 neurons, Output layer – 1017 neurons	ReLU Activation function	Adam optimizer, epochs = 200, Lr = adaptive learning rate	15 milli seconds	In terms of 10^{-5}

been mentioned that an extra care is needed to develop an inverse neural network and more training samples are required to train an inverse MLP model when comparing with the forward problem [98]. Most of the applications have been restricted to 1D or 2D problems and even if a study considers a 3D problem, that has also been restricted to simple geometry [82]. Also, it is worth noting that MLPs have been used to predict only a limited number of parameters as an output for each case. Most of the systems included 3 or 4 layers with two hidden layers except only one study has even considered 4 hidden layers for a FE modelling of a block [59]. However, there is no clear pathway to determine the architecture of a MLP model and it has been done with respect to a trial-and-error process in most of the studies [59]. The ReLU activation function is most commonly used as the activation function, and it has even provided a higher accuracy whilst the application of sigmoid function has also been identified another commonly used activation function. It is difficult to estimate the most optimum size of the dataset for a MLP network. However, it is understandable that the size of the dataset depends on the structure/model to be analyzed and also, it is possible to achieve a higher accuracy by increasing the size of the dataset. Febrianto et al. [31] attempted a bootstrap approach which develops multiple neural network models by considering randomly selected datasets to further ensure the accuracy of MLPs in predicting critical scenarios. It has been found that an ensemble of MLPs that provides the output of the model as the average or combination of the trained models performs better than considering one MLP that perform best. Most of the models reported a higher time conservation via MLPs (inference time is mostly less than a second for many cases) when compared to the actual FE modelling proving the ability of MLPs in generating real-time responses. Adam optimization algorithm can also be identified as the commonly used optimizer for MLPs and it has also been recommended as the most appropriate optimizer for ANNs [81]. Also, it is more common to see the mean squared error (MSE) or mean absolute error (MAE) as the loss function of most of the developed MLP models.

MLPs are attractive due to its simple straightforward architecture and several conceptual benefits. They have been known as universal approximators when the appropriate activation function has been utilized in the model development [89]. Also, they do not encounter the distribution of the given data and generalize well to produce more accurate assumptions even though the given dataset possesses a complex or any unknown distributions [40]. Hence, it has the capacity to be engaged in surrogation effectively. As well as the benefits, there are some inherent drawbacks of them when using as surrogates for FE modelling. MLPs create a relationship between the given set of inputs and outputs without considering the underlying physics of the problem or even any geometrical relationships among the given parameters. Consequently, even though the model trained with good accuracy, they may be considered as more statistical models. Hence, physics informed MLPs have been introduced more recently to overcome this problem and it will be discussed in a later chapter (Section 6.0) in this review. Also, it has been identified that MLPs poorly perform for complex FE meshes when there is a higher number of trainable parameters [65]. The time factor is not taken into consideration in MLPs, and it hinders the application of MLPs for time history analyses. However, Guan et al. [47] used an MLP for a time history analysis in a study where the constitutive properties of soils are determined by considering strain components of two consecutive times together as an input to the model whilst achieving a higher prediction accuracy of 0.6.

6.1.2. RNN (Recurrent neural Network)

A RNN (Recurrent Neural Network) model also generates an output sequence which corresponds to a given input sequence similar to MLPs. However, it employs a series of parameters that are related to the historical context of the data. This inherent feature equips the RNN with the ability to effectively manage and account for the dependencies on past information within the input sequence [43]. Also, it has the capacity to

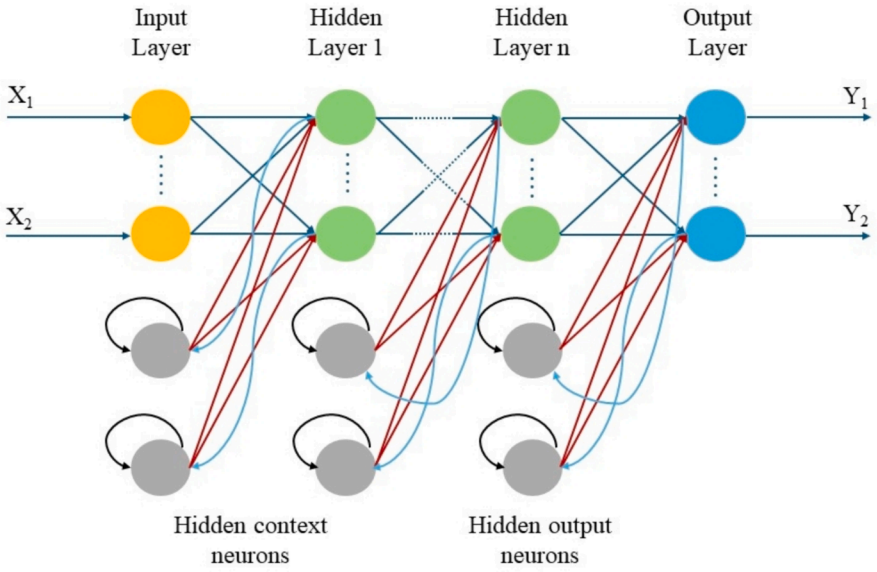


Fig. 7. Schematic of Recurrent Neural Network.

track the time-related patterns and forecast the future as well [124]. The most common applications of RNNs can be identified as Time series analysis, Speech recognition, Video analysis, Anomaly Detection etc. The schematic of the RNN can be presented as in Fig. 7 (X_1, X_2 – inputs and Y_1, Y_2 – outputs).

As depicted in Fig. 7, the RNN is an extended version of feed forward neural networks. The main difference is that it introduces a context layer that is appended to every hidden layer and the output layer. Within these layers, there are units known as context neurons, and these neurons are triggered into action by the output signals from their respective hidden or output neurons. During the learning process the connection strengths, also known as synaptic weights of the hidden and context neurons will be fine-tuned in such a way that to ensure when the network is given a particular input, its output closely matches the expected or target values [124]. Also, the number of hidden layers is a crucial factor of a RNN model and it directly affects its performance and accuracy [80]. Unlike the feed forward neural networks, communications between layers are two-way in RNNs. Usually, RNNs have the ability to retain information via context neurons, hence it does not generate the outputs relying only on the instant inputs but also consider trends in the input data sequence. Accordingly, these models yields to high accuracy when adopted in a time series or any sequential process [80].

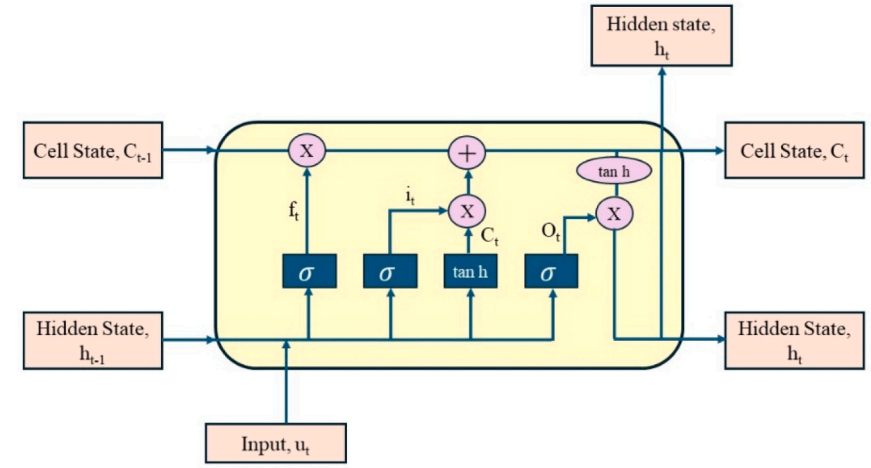
Within the realm of diverse RNN variants, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) have garnered significant popularity [46] where both the types have been implemented to overcome the vanishing gradient problem of general RNNs whilst handling long-term dependencies in sequential data. However, they are different in terms of the model architecture and the way of the data handling. They are equipped with gating mechanisms that facilitate the capacity to deal with long −term dependencies effectively. Table 4 compares the differences between those two structures whilst Fig. 8 (a) shows the architecture of LSTM (f_t – forget gate, i_t – input gate, O_t – output gate, s – sigmoid activation function) and Fig. 8 (b) shows the architecture of GRU. However, the decision upon which type should be used for the analysis totally depends on the purpose and sometimes their performances may not be significantly different with each other.

Table 4
Differences between LSTM and GRU networks.

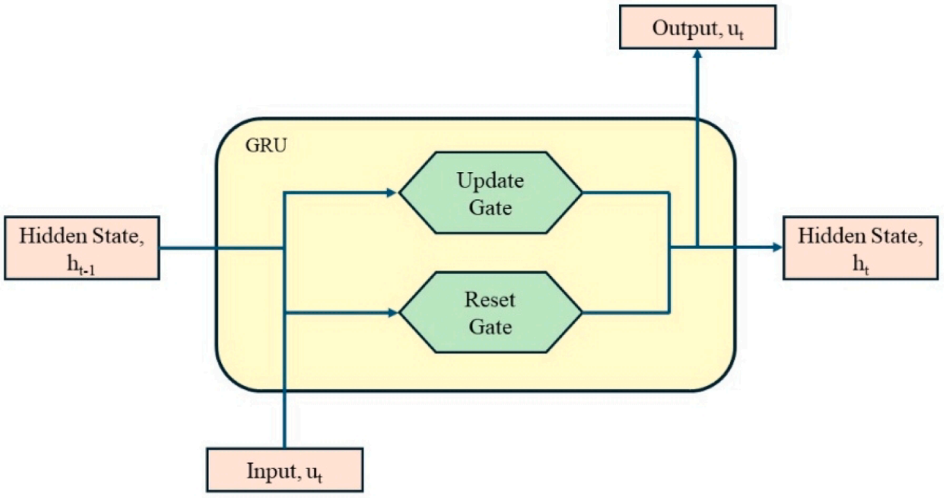
LSTM	GRU
Three gates in the architecture; input gate, forget gate, output gate	Two gates in the architecture; update gate, reset gate
More parameters due to the additional gate: forget gate	Fewer parameters
Can learn more complex patterns	Can learn more complex patterns
Slower training time	Faster training time

RNNs surrogate time history analyses of a FE analysis process. It has the capacity to retain information of the previous state, concerning it and generate the output of the current state. Hence, for any kind of FE analyses where the temporal variation is a concerning factor, RNNs can effectively serve as the surrogate for the analysis procedure [46]. Table 5 shows different applications of RNN for FE analysis of different types of structures in the civil engineering domain. It is worth noting that RNNs characteristic of leveraging output from the previous step as the input for the current step, makes them ideal for capturing the history dependence of materials and handling the non-linearity in such analyses [90]. It is an inherent characteristic of FE analysis that increasing the mesh resolution leads to improved results. However, a study by Ghavamian and Simone [43] aimed at creating a surrogate for the micro model in the multiscale finite element analysis of material models has shed light on the fact that RNNs operate independently of the FE mesh resolution and complexity of the micro model or the constitutive model. This means that the same RNN model can be effectively applied, whether the mesh is coarse or fine and even the complexity of the micro model or the constitutive material model stays at a higher level. Also, Ninić et al. [124] proved the practical application of RNNs for real-time FE simulations by using this technique in Wehrhahn-line metro project Düsseldorf, Germany. The unique characteristic of RNNs in mimicking the time series forecasting problems plays a considerable role when estimating dynamic responses of buildings under FEM.

RNNs are notorious for encountering the vanishing gradient issue. This problem becomes apparent when training RNN models using optimizers that rely on gradients, causing difficulties in effectively updating the parameters of the network during training [58]. The issue of vanishing gradients results in prolonged training times. It has been proposed to adopt a specialized cell known as Long-Short Term Memory (LSTM) to address this issue [43]. Also, RNNs tend to overfit the training data easily as it closely fits with the training data due to its high generalizability. Regularization methods are the ideal solutions for this and amongst, the dropout method that randomly dropping out nodes to form diverse architectures for the RNN model is highly popular [80].



(a) LSTM Network



(b) GRU Network

Fig. 8. Architecture of (a) LSTM (b) GRU.

6.1.3. CNN (Convolutional neural Network)

The utilization of convolutional neural networks has emerged as a widely discussed subject in forecasting complex physical systems and also, it is regarded as the fundamental element in addressing prediction challenges within the realm of physics [34]. CNN is a type of neural network that primarily works with array-based data and is often used for applications such as image analysis or identification processes. In fact, an image represents a combination of three coloured 2D-arrays [123]. It has the capacity to extract features from images upon training [136]. The concept of CNNs were first introduced by Lecun et al. [87] for the purpose of classifying the characters which were handwritten. Afterwards, CNNs evolved for the past few decades and nowadays it has gained huge popularity. Mainly, CNNs are comprised of two components: convolutional layer and pooling layer. Features of an image are detected using filters in the convolutional layer and a function like ReLU is being used here to introduce a non-linearity into the model. Subsequently, a pooling layer comes into action and reduces the dimensions of arrays [123] by following different criteria such as categorizing similar features together or averaging or obtaining the maximum value. Pooling is beneficial as it reduces computational complexity, reduces the possibility of overfitting and tolerating variations or distortions of the model. Accordingly, the CNN can consist of a number of convolutions with ReLU layers and pooling layers whilst flattening at the end and passing through a dense neural network. Apart from the standard CNN architecture, the most popular architecture of CNNs in surrogate modelling of FEM is the CNN U-Net, named due to its U-shaped architecture, See Fig. 9. The shown architecture has 5 encoder blocks (E1..,E5) and 5 decoder blocks (D1…, D5). Encoders process the input and extract salient features with the use of series of convolutional and pooling layers. Then, whilst accommodating transposed convolution operation and incorporating skip connections, decoders up-sample the features. Skip connections assist in passing immediate layers and transferring the output into subsequent layers. The decoder then up samples the features using transverse convolution and concatenates them with the process called skip connections. This allows for the direct transfer of essential features from encoder to decoder to produce more accurate results at the end.

Recently, the use of convolutional neural networks has become a hot topic in predicting high-dimensional physical systems since image representations can be used for 2D FE meshes and the stress can be predicted with the use of feature detection ability of CNNs. Upon adding different modifications to the U-Net architecture, several other CNN architectures have been derived and have been used for simulating the mechanical systems. Table 6 describes that kind of architecture that was

Table 5
Applications of RNNs in surrogating FE modelling.

Author	Purpose	Size of the Data set	Data Preparation	Structure of the ML model	Activation Function	Model Training	Inference time	Performance
Ninić et al. [124]	To support tunnel boring process via real-time simulations and monitoring based predictions	Training – 100 samples, Validation – 50 samples	Using object-oriented FE framework KRATOS	Three layers with one hidden layer that includes 20 context neurons	Sigmoid function	"Batch mode" learning process was used by minimizing the mean squared error.	Used only seconds to few minutes	RMSE of 3.1 % for training and 3.5 % for validation
Freitag et al. [33]	To forecast the time-dependent settlement patterns caused by mechanized tunneling.	Values of different parameters were changed in a range of values to define data set. Size is not mentioned.	Using object-oriented FE framework KRATOS	Four layers with two hidden layers that includes 10 hidden neurons	Tanh for hidden neurons and linear function for output neurons	Levenberg-Marquardt (LM) backpropagation algorithm was used for training	4 s	The prediction error was 6.2 % and 8.9 % for the lower and upper bound respectively
Ghavamian and Simone [43]	To develop a surrogate for the micro model in a multiscale finite element analysis	Formed by pulling out a bar a different displacement rates until it reaches 2 mm displacement	FEM simulation on a set of macro models with desired configurations and obtaining macro strain and stress	Consists of nodes (mLSTM, dropout, dense cells) for operations and edges including strain, stress and historical response vectors	Not mentioned	Trained using Adam optimization algorithm until the mean squared error is sufficiently small	Reported that the RNN is remarkably faster than FEM	Minimum error is not mentioned. Predicted values and results of the micro model have been compared using graphs include reaction forces at one end of the bar
Wu et al. [164]	To surrogate *meso*-scale boundary value problem under multi-scale simulations	9000 sequential data was generated and 30 % of it were used for training	2D multi-scale FE analysis	GRU architecture has been selected. Two feedforward NNs with one hidden layer of 60 neurons for one and two hidden layers of 100 neurons for the other one.	Leaky ReLU function	Omitted from the paper and followed the recommendations of PyTorch library	Reduced by 4 orders	MSE nearly equals to 4.3×10^{-6} for data having length of 200 and 1.5×10^{-5} for data having length of 1000
Kim [80]	To estimate the seismic response of 11-storey and 27-storey buildings under semi-active control systems	13 samples, 10 for testing and 3 for verification	Non-linear FE analysis using FE software and using Bouc-Wen model in MATLAB	LSTM was used with 5 hidden layers for the default model	Tanh function	Adam optimization was used with a learning rate of 0.01 and trained for 1000 epochs	0.013 s for 11-storey and 0.0142 s for 27-storey	1.232×10^{-3}, 1.103×10^{-3} and 5.249×10^{-3} of RMSE for three tested scenarios
Li and Zhuang [90]	To simulate multi-scale constitution	200 samples, 120 – training, 40 – testing and 40 – validation	Using multi-level finite element method (FE^2) in ABAQUS	GRU architecture has been selected	Tanh function	Adam optimization was used with learning rate of 0.001, dropout rate 0.95	Nearly 4 s in all tested scenarios	Maximum accuracy of 96.7 % in prediction
Cao et al. [19]	To assess building damage categories caused by soil-structure interaction in mechanized tunneling	60 samples, 80 % – training and 20 % – for final validation, out of 80 % of training data 70 %, 15 % and 15 % for training, validation and testing	Using object-oriented FE framework KRATOS	Four layers with two hidden layers each with 20 neurons	Tan h for hidden neurons and linear function for output neurons	Levenberg-Marquardt (LM) algorithm was used for training	Less than 1 s	Coefficient of determination R^2 of 0.99
Greve and van de Weg [46]	To develop parametric surrogate models for finite element simulations that are parametrized	10 samples with train to test split 9:1	Using explicit crash FE solver	Standard 4-gated LSTM cell. Tested for two models considering node by node and all nodes at once	Sigmoid function	Adam optimizer is used with the learning rate or 0.001 and gaussian dropout with a probability of rate 0.1.	Less than a second for all approaches. 0.045 s is the maximum.	R^2 value was higher than 0.9 for all models and the minimum one was 0.9268.
Guan et al. [47]	To predict the constitutive behavior of soils	128 datasets with 0.1 validation split	2D plane stress analysis disregarding the gravity	LSTM with 2 hidden layers that includes 128 hidden neurons, 3 neurons in both input and output layers	Default Activation function	Adam optimizer, epochs = 1000, Lr = 0.001, batch size – 128	Significant acceleration was observed in the calculation time	MAE of 0.56

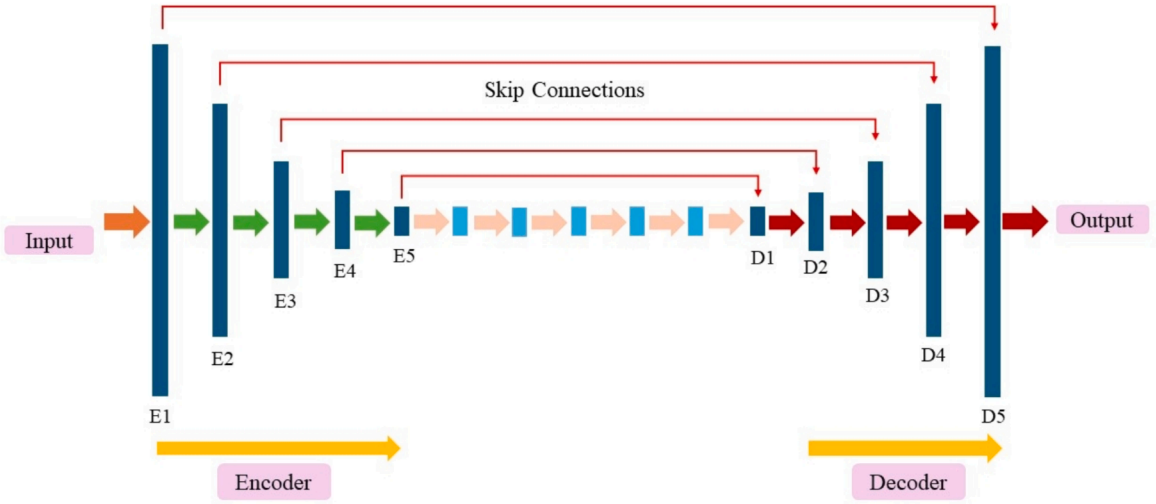


Fig. 9. U-Net architecture of CNN model.

Table 6
Features of different modifications added to CNNs.

CNN Architecture	Features
Standard U-Net of CNNs	– U-Net architecture – Consists of convolution + ReLU and pooling layers – The most common architecture of CNNs in most cases
ResNet	– Residual blocks are being used to address vanishing gradient problem. – Model complexity will slightly increase.
Inception Blocks	– Utilizes multi-size blocks. – Enhances the ability of representing complex patterns. – Makes the CNN wider not deeper and reduces the computational cost
SE-Res Net	– Squeeze and excitations blocks will be added to the ResNet – Model channel-wise dependencies. – Focuses on more relevant feature supressing others

commonly used in the field of civil engineering for the purposes of structural analyses.

Table 7 presents several recent applications of CNNs in surrogating FE modelling. It is highly notable that most of the applications of CNNs are predicting stresses and it has been achieved a higher accuracy in surrogating the stress predictions via FEM. Few studies have mentioned that the inference time has been substantially accelerated by CNNs [147,136] even though some studies have not considered about the time conservation as much as reducing the computational complexity of the analysis process. Further, it can be identified that almost all the studies followed the general U-Net architecture except some studies used residual blocks, inception blocks, squeeze and excitation blocks in order to obtain a higher computational efficiency. In a study of Fung [38] to use the CNNs to solve torsion problem of an elastic bar with arbitrary cross section, they tried different CNN architectures and found that DenseNet201 [62] with 201 layers including transition layers which uses a dense connectivity pattern where each layer has been connected with every other layer and InceptionResNetV2 [149]which combines inception modules with residual connections with the purpose of capturing multi-scale features whilst preventing vanishing gradient issue models performed the best. The Leaky ReLU can be identified as the most commonly used activation function whilst in some applications the use of Tanh and sigmoid activation functions together with ReLU have also been identified. It was noticeable that the Adam optimizer has been used for the training of most of the models. Apart from other ANN models, in CNNs the SSIM (Structural Similarity Index Measure) can be used to evaluate the accuracy of the trained model as it checks the similarity in between the actual images and tested images [147]. Also, in a study that was conducted to predict the failure patterns of composites using CNNs, Sepasdar et al. [143] highlighted the application of attention loss as the loss function which is a modified version of the mean absolute error (MAE) as it is capable of escalating the absolute error of pixels that are more critical in identifying crack patterns of materials. It has been also identified that CNNs can effectively surrogate higher-dimensional spatial problems when comparing with the usual feed forward neural network models [61].

Despite the widespread utilization of CNNs as substitutes for finite element analysis, they come with notable limitations. First, dealing with complex FEA scenarios poses a challenge in input tensor encoding due to its requirement of fixed input tensor sizes. Also, given that CNNs are confined to process calculations within Euclidean coordinates, many surrogate models relying on CNNs opt for meshes of square or cube shapes. These meshes necessitate the creation of an uniform grid on the input FE model [34]. In this context, there is an option to integrate the mesh within a standardized domain and apply interpolation. Nevertheless, such a method could lead to substantial interpolation inaccuracies when handling intricate geometric models and might impose restrictions on required mesh refinements [30]. CNNs inherently lack the ability to handle variations in rotation and scale. Furthermore, due to the predominantly unstructured nature of FE models, the input tensor of CNNs struggles to seamlessly integrate with the FE meshes [34]. With these drawbacks and the considerable success of CNNs with Euclidean data, many researchers have strived to establish convolution methods for non-Euclidean data [113]. However, they are associated with substantial pre-processing cost [28] and in that case Graph Neural Networks (GNNs) have introduced as an ideal alternative to CNNs. In a study, Rezasefat and Hogan [136] attempted to predict the stress variation of finite plates with inclusions and tried to evaluate the accuracy of CNN

Table 7
Applications of CNNs in surrogating FE modelling.

Author	Purpose	Size of the Data set	Data Preparation	Structure of the ML model	Activation Function	Model Training	Inference time	Performance
Nie et al. [123]	To predict the stress variation in a 2D cantilevered beam	100,000 test and 20,960 training data	2D FE modelling using SolidsPy	SCSNet – one input channel with two convolutional and pooling layers StressNet – multiple input channels with three convolutional and pooling layers	ReLU function	Adam optimizer with exponentially reducing learning rate, batch size – 256,	1.56 s to render 120,960 samples 10.4 s to render 120,960 samples	MRE – 10.43 % MRE – 2.04 %
Yang et al. [165]	To obtain the stress/strain curve of composites	100,000 data with a 95 % test and 5 % train split	The crack phase field solver, designed for quasi-static fracture of elastic solids, was implemented in ABAQUS using a user-element subroutine (UEL) based on a hybrid formulation.	All CNN layers were with 11 filters, stride of 1 and batch normalization. 2D max pooling was used for last 9 convolutional layers avoiding the first three.	Leaky ReLU function	Adam optimizer with 30 epochs, batch size – 16	Mentioned as lesser time than FEM	Less than 10 % error for all tested parameters
Bolandi et al. [13]	To predict the von-mises stress distribution in loaded steel plates	83,558 training data and 20,890 testing data (80 %, 20 %)	PDE toolbox in MATLAB was used to generate outputs	U-Net based architecture – 7 encoder blocks, 6 – decoder blocks, 3 – SE ResNet layers, 1 – Efficient net layer	ReLU and Sigmoid functions	Adam optimizer with learning rates of 1e-3, 1e-6 and 1e-5, batch size 16,	Not mentioned	Mean absolute error of 0.9 % and peak error 0.49 %
Sepasdar et al. [143]	To predict the stress and failure patterns of composite materials (Two CNNs were used)	4000 training and 500 validation data	2D FE modelling of microstructural representations under transverse tension	A modified U-Net architecture – 5 blocks for both encoder and decoder each with kernel size of 4 and stride 2	Leaky ReLU function	Adam optimizer with 120 epochs, batch size – 5, the output of the 1st network was used as the input for the 2nd network for training	Not mentioned	Accuracy for MAE as loss function – 85 %, for attention objective function – 90 %
Ibragimova et al. [65]	To predict deformation of metals under uniaxial loading	12,000 samples, 10,000 – training, 1000 – validation and 1000 – testing	2D crystal plasticity FE method	Modified U-Net architecture – 5 blocks for encoder, 5 blocks for decoder, 3x3 filters including batch normalisation layers	Tanh and ReLU functions	Adam optimizer, number of epochs – 100, batch size – 4, initial learning rate – 10^{-3}	56 GPU milliseconds	RMSE of testing as 6 x 10^{-4}, validation error – 5.8 x 10^{-4}
Hsiao et al. [61]	To predict factor of safety and slope failure	In different cases 10 % to 16 % data were used for training and the rest was used for testing	Plane stress analysis using PLAXIS software	4-layer model with 40 filters in each layer of the model	Not mentioned	Mean Absolute Percentage error has been used as the loss function	Reduction of computation time was reported	MAE of testing was 1.16 x 10^{-2}
Shokrollahi et al. [147]	To predict the stress fields of composite materials with uniform, non-uniform and free-form fiber sets	1400 images after 4-fold flipping with 85 % training and 15 % testing	Plain strain 2D FEM analysis	Specialised U-Net architecture with 6 repetitive blocks in the encoder each with 2x2 max pooling operation. Updated this with two deep layers with 32 kernels at the start and end.	ReLU function	Training was accomplished by minimizing weighted score mean error as the loss function	0.5 s CPU and 0.05 s GPU time	Structural similarity index 0.977 and MSE of 0.0009
Rezasefat and Hogan [136]	To predict the stress distribution and concentrations around defected areas of a plate	900 training data and 1oo testing data	2D FE modelling using Abaqus for finite plates with defects under plane stress and except body forces (linear elastic)	U-Net based model – 6 blocks in the encoder with 2x2 max pooling operation, two consecutive 3x3 convolutions and six blocks in the decoder with 3x3 convolution in each	Leaky ReLU function	Adam optimizer with learning rate – 0.001, beta_01 – 0.9 and beta_02 – 0.09, batch size 32, trained for 200 epochs and the loss function was MSE	1.2 s for 100 samples	For training and testing the MSE was 0.086 and 0.171 while correlation was and 0.9740 and 0.9527 respectively
He et al. [56]	To predict the failure mechanism of a footing	12,332 data for 5 soil types, 70 % training, 15 % testing and	2D FE modelling with an implicit dynamic scheme of footings	Modified U-Net architecture – 5 blocks for encoder and 4 blocks for decoder, 3x3 filters and 2x2 stride	Leaky ReLU and ReLU functions	Trained until the weighted error gets minimum by considering it as a loss function	Not mentioned	3.3 % error for test data

(*continued on next page*)

Table 7 (*continued*)

Author	Purpose	Size of the Data set	Data Preparation	Structure of the ML model	Activation Function	Model Training	Inference time	Performance
Khorrami et al. [79]	To predict the stress in visco-plastic materials	15 % validation 1000 samples, 80 % – training and 20 % – testing	Using the boundary value problem for quasi-static mechanical equilibrium	U-Net architecture with 32 filters, kernel size of 9 with max pooling	ReLU and Sigmoid functions	Adam optimizer was used with Lr = 0.001, loss function was MAE	500 times faster (0.15 s)	MAE for testing 1.265 as the minimum

model by changing different parameters of the dataset. Their conclusions verified that the size of the dataset and the sensitivity of the dataset in terms of the resolution positively impact the accuracy of the model. Also, it has been found that the extrapolation capacity of CNNs is limited and can lead to significantly deviated results with an error of 64 % (maximum) if the input data is beyond the limits of the training data sample. Hence, it is clear that the generalization capacity of CNNs is also confined to the limits of the training data sample. In FEM, several parameters including geometry, boundary conditions, loadings etc. are required to be entered as an input to the model and in such cases, it is recommended to use different input images for each parameter to achieve a higher accuracy [13,123].

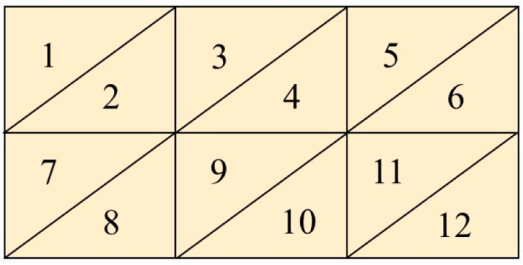
6.1.4. GNN (Graph neural Network)

Scarselli et al. [142] proposed the concept of GNNs for the first time, enabling the processing of data specifically structured within graph domains. After that, GNNs were identified as one of the types of neural networks that directly process non-Euclidean data unlike handling vectorized, or image-based data and they have been extensively used in various fields such as knowledge graphs, social sciences, recommendation systems, medicinal sciences, molecular structures, traffic forecasting etc. As the name implies, Graph Neural Networks (GNNs) conduct computations that traverse and manipulate data structures known as graphs. Gulakala et al. [49] mentioned three different graph structures: Directed/Undirected, Homogeneous/Heterogeneous and Static/Dynamic. In the meanwhile, it is an inherent characteristic of the FE analysis to employ a mesh-based modelling sequence that represents a discretized version of a structure including only nodes and different types of elements, when solving underlying partial differential equations of a particular system. By naturally extending mesh data into a graphical format, graph neural networks gain all the benefits associated with utilizing interconnected domains and the input parameters will be considered as different features of the graph [146]. GNNs offer few major advantages over CNNs when working as a surrogate for FE modelling. In fact, GNNs enhance refinement of meshes in proximity to stress/strain concentrators such as defects, material discontinuities, irregularities in the surface of a particular structure while allowing to obtain more precise localized predictions and substantially reduced computational cost. Further, unstructured mesh-oriented models enable the acquisition of the behavior of a system solitarily disregarding the resolution and this implies that various mesh sizes can be employed during runtime without hindering the learning process. Also, GNNs offer a more efficient means of representing architected truss metamodels [109].

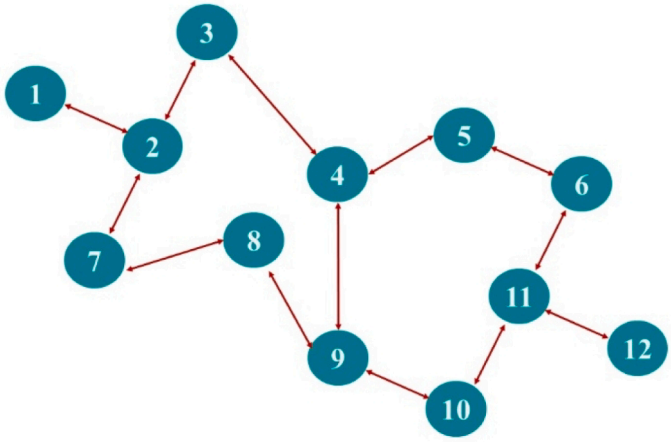
Graphs are often depicted as a data structure denoted by $G = (V,E)$, wherein V signifies vertices, and E represents the edges that establish connections between pairs of those vertices. The relationships between vertices of a graph via edges can be represented in an adjacency matrix. If the number of vertices is n, the order of adjacency matrix (A) will be nxn and $A_{ij} = 1$ if an edge exists in between two vertices whilst $A_{ij} =$ 0otherwise. Fig. 10 shows a graph representation of an FE mesh and its adjacency matrix. The conversion of FEM to graph structures is fundamental to applying GNNs as FEM surrogates. This process involves defining nodes and edges to represent FEM meshes and assigning features that capture the structural and material characteristics of the model. The FE mesh will be depicted as an undirected graph in order to utilize the concept of GNNs. Based on that, the prediction tasks of a GNN for FEM can be performed under three levels: Node-level, Edge-level and Graph-level [126]. In the graph representation of FEM, nodes correspond to the key entities in the FEM mesh while edges represent the connections between nodes. For instance, the nodes in an FE model of a truss structure represent the nodal joints, and their features may include nodal coordinates, boundary conditions and applied nodes while edges are defined by bars connecting the joints, with features including their material properties and geometric dimensions [126]. For 2D or 3D mesh, the elements themselves are often treated as nodes, with features such as material properties, element centroid coordinates and associated stress and strain values while edges may represent the adjacency between elements and usually, edges indicate shared boundaries or interfaces between adjacent elements [69]. In the adjacency matrix, 1 represents a connection whilst 0 represents the opposite. When defining the FE mesh as a GNN, the connections among the elements are very important and after the FE mesh will be represented as a graph embedding, it will be encoded, messages will be passed in between graph nodes and decoded at the end with the results of the analysis process. This representation enables GNNs to process spatially distributed data efficiently, leveraging the inherent connectivity of the FE mesh. GNNs have the capacity to capture localized stress/strain distributions and refine predictions near critical regions, such as stress concentrators or defects by employing node and edge features to the graph structure.

Table 8 shows a summary of application of GNNs in the civil engineering domain and presents how the GNN models have been implemented depending on the intended use. According to Table 7, it is clear that the computational time of FE modelling has been drastically reduced in almost all the instances whilst also keeping the prediction accuracy at a very high level. It verifies the potential of using GNNs as a surrogate for FE modelling. Another common feature of the developed GNNs was the undirected graph structure to represent the FE mesh. Also, a message passing framework has also been used in most cases to accurately simulate the model. It was observed that nodes and elements of truss structures were considered as same nodes and elements of the graph structure whilst for 2D or 3D meshes the elements of the mesh has been considered as nodes and connections in between elements have been considered as elements. Hence, representing truss bridges as GNNs is much simpler than other 2D or 3D meshes. PyTorch Geometric seems to be the most popular library to implement the GNNs. ReLU function was most commonly used as activation function in GNNs whilst the mean squared error has been used as the loss function for most of the cases.

However, several challenges and limitations have also been noticed. In a study to predict stress, strain and deformed shapes in material and structural systems using GNNs, Maurizi et al. [109] presented that GNNs can grasp loading conditions that link deformed shapes with stress fields, while also capturing the inherent relationships between the structure, stress/strain and deformation fields. Also, he emphasized the possibility of accelerating computation time of FEM while stating the memory-demanding nature and the lack of generalizability as some limitations of the trained GNNs. Also, Gulakala et al. [49] provided evidences on another constraint of GNNs via a numerical study on its



(a) FE Mesh



(b) Graph Embedding

	1	2	3	4	5	6	7	8	9	10	11	12
1	0	1	0	0	0	0	0	0	0	0	0	0
2	1	0	1	0	0	0	1	0	0	0	0	0
3	0	1	0	1	0	0	0	0	0	0	0	0
4	0	0	1	0	1	0	0	0	1	0	0	0
5	0	0	0	1	0	1	0	0	0	0	0	0
6	0	0	0	0	1	0	0	0	0	0	1	0
7	0	1	0	0	0	0	0	1	0	0	0	0
8	0	0	0	0	0	0	1	0	1	0	0	0
9	0	0	0	1	0	0	0	1	0	1	0	0
10	0	0	0	0	0	0	0	0	1	0	1	0
11	0	0	0	0	0	1	0	0	0	1	0	1
12	0	0	0	0	0	0	0	0	0	0	1	0

(c) Adjacency Matrix

Fig. 10. Surrogate model architecture of GNNs.

Table 8
Applications of GNNs in surrogating FE modelling.

Author	Purpose	Size of the Data set	Data Preparation	Structure of the ML model	Activation Function	Model Training	Inference time	Performance
Maurizi et al. [109]	To predict Stress, strain fields and deformed shapes of, - Unidirectional fiber composites - Wrinkled interfacial layers - Lattice structures	90 % training and 10 % testing 500 500 762	Using Abaqus under 2D static loading	PyTorch Geometric – Encoder, Message Passing and Decoder – used two neural networks	ReLU function	MAE as loss function and the Adam optimizer – to avoid overfitting L2 regularization technique with weight decay of 5 X 10^{-4}. Initial Lr = 0.01 with exponential decay = 0.9	Time required for each step; 0.015 s 0.016 s 0.009 s	Average Mean Absolute error for three models, 0.07 0.22 0.29
Shivaditya et al. [146]	To design the metal forging process	75 % training and 25 % testing 40	FEM simulations of a Yoke metal forging process	PyTorch Geometric – takes a graph as input and returns graph as output	ReLU function	MAE as the loss function – trained for 2500 epochs with a method for stochastic optimization using Lr = 8 X 10^{-4}	99.9 %	Average coefficient of determination is 93.7 % for lower die and 92.8 % for upper die
Gulakala et al. [49]	To analyse the stress of a plate with a hole under different loading conditions	92.5 % training and 7.5 % testing 540	Aluminum plate with elastoplastic behaviour using the isotropic Johnson-Cook model using ABAQUS	PyTorch Geometric – Encoder-Decoder architecture	Not mentioned	MSE as the loss function and Adam optimizer, Lr = 3 X 10^{-3} with an exponential decay over epochs starting from 200th epoch. Trained for 1000 epochs	Not mentioned	Mean squared error of 5.89 X 10^{-2} for the whole test batch
Fu et al. [34]	Cantilever beam problem for, - Predicting stress and - Topology optimization	70 % training, 30 % for testing and validation	ABAQUS – 2D analysis and density-based simple isotropic material with penalization method to optimize the cantilever beam structure	PyTorch Geometric – Boundary oriented graph embedding approach – 3 later DeepGCN model	Not mentioned	MSE as the loss function and Adam optimization algorithm, Lr = 0.01, 500 epochs, has done for both 3-layer and 8-layer models	Consumed only 10 ms	MSE of 0.011706 for stress field prediction and 0.002735 for topological optimization for 3-layer model
Jiang and Chen [69]	Von mises stress prediction of a simply supported rectangular metal sheet under three cases - Varying magnitude of the load - Varying point of application of the load - Varying geometrical parameters	Around 94 % for training and 6 % for validation 136 136 8192	FE analysis based on isotropic and homogeneous material	PyTorch Geometric – encoder-decoder architecture with one and three fully connected graph convolutional layers for message passing between nodes	ReLU function	RWSLE was the loss function and Adam optimizer with MultiStepLR scheduler, Lr = 0.0001	Speed up the simulation by 316 times on average	MAPE for three cases are as follows: 0.34 % 0.43 % 0.61 %
Nourian et al. [126]	Predicting nodal displacements of different truss models - 10-bar planar truss - 72-bar space truss - 200-bar planar truss	Trained using all data samples 1000	Simple finite element model to determine nodal displacements	Spektral library – encode-process-decode architecture	ReLU function	MSE was the loss function and Adam optimizer, Lr = 0.01, exponential decay at rate of 0.5 over 500 steps.	Reported a reduction in the computational time	MSE for each truss model 0.022 0.019 0.496
Taghizadeh et al. [150]	Predicting stress for following scenarios	Training/ testing split is not mentioned. Tested for low fidelity	FE analysis using PyAnsys software package	The best performing model included a combination of LFGNN and HFGNN. The LFGNN has 12 Graph Network (GN) blocks, each with a node MLP (64 hidden units, 2	Not mentioned	Adam optimization algorithm was used and trained for 500 epochs with a learning	Minimum of 0.0136 s	rates ranging from 1.28 % to 1.93 %

(*continued on next page*)

Table 8 (*continued*)

Author	Purpose	Size of the Data set	Data Preparation	Structure of the ML model	Activation Function	Model Training	Inference time	Performance
		and high fidelity meshes		layers) and edge MLP (64 hidden units, 2 layers). The HFGNN uses 3 EdgeConv		rate of 0.0005 with an		
	- 2D plate with notches	5000 low-fidelity and 500 high-fidelity data points		layers, each with an MLP (64 hidden units, 2 layers). Both LFGNN and HFGNN employ encoder–processor–decoder structures		exponential decay of 0.99985 per one epoch.		
- 2D plate with a hole								

extrapolation ability and revealed that it limits at 30 % over the training boundaries, beyond which their performance notably declines. Furthermore, extensively layering GNNs can lead to over-smoothing challenges, resulting in graph-vertex embeddings becoming indistinguishable and ultimately rendering GNN ineffective in generating predictions [91]. Consequently, the preference has leaned towards employing a 3-layer GNN, characterized by its focus on shorter ranges of interactions among graph vertices and constrained message-passing distances [34]whilst GNNs are performing well for graphs with small number of vertices or graphs with large number of vertices with short-range interactions.

6.2. PINN (Physics-informed neural Networks) as a framework enhancing FEMs

The use of Physics-Informed Neural Networks (PINNs) as a framework for integrating physical laws into neural network training is explored in this section, highlighting their role in enhancing FEM simulations.

6.2.1. Overview of PINNs

Conventional neural networks map in between the input and output data of them whilst considering the required information to the analysis process as inputs and results from the analysis process as outputs when surrogating the FE analysis [67]. Hence, if a neural network is properly trained, the surrogation process can be carried out with reasonable accuracy. Even though the representation of data and the resulting accuracy may vary across different neural network architecture, this is the general approach followed during the surrogation process. Hence, the underlying physics of the interesting structure is not taken into account and when the training dataset does not encompass the necessary range of data, the generalization capacity of conventional models is diminished [86]. In order to address those issues whilst expecting better generalization and scalability, physics informed neural networks (PINNs) have been introduced. PINNs can be identified as a combination of data-driven and physics-based model development techniques where the traditional neural networks are modified by incorporating the fundamental physics of a system [5].

Table 9
Comparison of collocation and energy-based loss functions.

Collocation loss function	Energy-based loss function
Minimize the residual of the PDE at each collocation point	Minimizes the potential energy ensuring the equilibrium state
Directly enforces PDE and boundary conditions at all sample points	Ensures the balance of loss terms as they present potential energies
Poor generalization may happen due to scale of loss terms, specifically due to boundary conditions	High displacement accuracy, but lower stress accuracy due to indirect use of equilibrium and boundary conditions in the loss function
Computationally expensive due to scaling issues	Computationally efficient for displacement predictions whilst stress predictions need a post processing
Recommended for problems where precise enforcement of PDEs and boundary conditions are critical	Recommended for problems that do the displacement predictions mainly

6.2.2. Techniques for embedding physics in PINNs

PINNs can be in the form of a general feedforward network or any other neural network architecture [111]. Usually, the physical aspects of the system are embedded into PINNs to the loss function in the form of a partial differential equation (PDE) where the underlying physics is acting as a constraint in the optimization process and due to that, their training process may require more time compared to that of conventional neural networks. Han et al. [55] categorized PINNs as a type of reinforcement learning as well. All the input and output parameters can be identified as in a normal feedforward neural network. However, the outputs will be considered as main parameters in partial differential equations that govern the structural system and form a loss function for further optimization. Bai et al. [5] identified two primary forms of loss functions: collocation loss functions and energy-based loss functions.

Table 9 compares the features of those two loss functions, that are commonly used in PINNs to embed the physical principles. Collocation loss functions enforce the PDEs and boundary conditions directly at collocation points, ensuring strict compliance but facing challenges with generalization and computational cost due to scaling issues. It is ideal for problems requiring precise PDE enforcement. In contrast, energy-based loss functions minimize the system's potential energy to ensure equilibrium, offering computational efficiency for displacement predictions. However, stress accuracy often requires post-processing. This method is better for problems focused on displacement predictions rather than stress analysis.

However, some PINNs utilize physics-based principles not only in the loss function but also in other aspects of their implementation. Table 10 summarizes findings from the literature review, highlighting studies that applied physics-based principles and the neural network architectures used in each case.

Table 10 predominantly highlights the use of MLPs as the neural network architecture, along with specialized approaches such as RNNs and Gaussian Process models when developing PINNs. The techniques to embed physics into PINNs are diverse and problem-specific, including the incorporation of governing equations (e.g., equilibrium equations, momentum balance, and Laplace's equation) and boundary conditions directly into the loss functions. Advanced methodologies such as weak and strong form formulations, residual-based FEA losses, and LSWR losses are employed to ensure that neural networks adhere to the underlying physics of the problem. Additional techniques like tensor differentiation for accurate derivative computation, domain decomposition for handling discontinuities, adaptive loss weighting, and transfer learning for improved computational efficiency further enhance the effectiveness of PINNs. These approaches not only improve the convergence and stability of the models but also allow them to address complex challenges, such as material heterogeneity, strain field discontinuities, and multi-variable optimization.

Table 10
Neural Network Architectures and Physics-embedding techniques in PINNs.

Reference	NN Architecture	Techniques to embed physics
Zhang et al. [171]	RNN (LSTM)	• Governing equations of motion, state dependency, and hysteretic constitutive relationships embedded in loss function • Tensor differential for derivative calculations • Latent variables for hysteretic dynamics
Haghighat et al. [53]	MLP	• Loss function incorporating momentum balance, constitutive relations and kinematic equations • Automatic differentiation for accurate derivate computation • Independent networks for displacements and stresses to improve optimization and accuracy
Huzni et al. [64]	Gaussian Process (GP) Surrogate model	• Governing equations of corrosion (Laplace's equation) embedded as boundary conditions • Gaussian process regression to emulate FEM • Functional boundary conditions from experimental data
Rezaei et al. [135]	MLP	• Energy-based loss function combining weak and strong forms • Separated networks for outputs, automatic differentiation, and connection loss terms enforcing constitutive relationships
Le-Duc et al. [86]	MLP	• Weak form governing equations embedded as residual-based FEA loss function. • Boundary conditions integrated into stiffness matrix and vector constraints
Bai et al. [5]	MLP	• LSWR loss function integrating equilibrium equations, boundary conditions and dimensionless scaling
Bastek and Kochmann [9]	MLP	• Weak form PDEs incorporated in the loss function • Variational principle minimized for training
Vahab et al. [160]	MLP	• Physics-based loss functions incorporating equilibrium equations, boundary conditions, and material interface compatibility • Domain decomposition for strain field discontinuities
Chen et al. [21]	MLP	• Governing differential equations and boundary conditions incorporated into the loss function • Adaptive loss weight control algorithm, and transfer learning for computational efficiency
Wang et al. [162]	MLP	• Loss function integrates FEM reconstruction with boundary and governing equation losses for enhanced convergence and stability • FE data distribution constraints used as a regular term.

6.2.3. *Application of PINNs for surrogating FEM*

Table 11 shows a summary of few studies that used PINNs as a surrogate for finite element modelling. Karniadakis et al. [76] highlighted that PINNs excel in solving multi-scale problems and inverse problems although for the forward or well-posed problems the conventional NNs may surpass the performance of PINNs as in forward problems, the relationships are well defined, and they often do not necessitate the integration of physics-based constraints and any additional data. PINNs often require a small set of data and more generalizable than conventional NNs due to its capability of conducting both interpolation and extrapolation processes [76,53]. In a parametric study of Haghighat et al. [53] on PINNs, it was observed that the efficiency of PINNs will be enhanced if separate networks are utilized for each individual variable (output) in the problem and the efficiency of PINNs was proved in analyzing both linear and non-linear problems. Since this study is a parametric investigation, Table 9 shows only the parameters of the most optimum model. Different numbers of hidden layers were also incorporated in this study, and it was clear that deeper models reduce the accuracy of them due to overfitting. As the loss function incorporates partial derivatives, the hyperbolic tangent function outperform the ReLU activation function in PINNs as ReLU forms discontinuous derivatives [53]. Le-Duc et al. [86] also notices a similar performance in hyperbolic tangent function. Zhang et al. [171] used a PINN that was based on multi-LSTM networks and compared its results with the classical LSTM network. It was noted that the physics-based LSTM is performing much better than the traditional network. The findings of Le-Duc et al. [86] also correlates with Zhang et al. [171] as the performance of PINNs based on MLPs performed better than conventional MLPs. Meethal et al. [111] also used a PINN and proved its efficiency in simulating both forward and inverse analysis processes in FEM. Wang et al. [162] introduced a novel addition to PINNs considering domain meshing and FEM reconstruction which showed a better performance than regular PINNs. Also, in a study by Zhang et al. [172] which has been conducted to solve differential equations of motion for forced vibrations in damped systems, it has been identified that for PINNs, selecting an appropriate neural network structure is critical for achieving accurate results. It recommends using a network with four to five layers, as fewer layers may lead to insufficiently modelled responses, while an excessive number of layers can increase computational complexity and optimization challenges. By carefully tuning network parameters, the approach shows potential for generalization to more complex structural analyses.

Even though PINNs perform better than NN models that use regular loss functions, their performance becomes limited when dealing with a slightly complex FE mesh. Soft regularization is not recommended for PINNs as they make the problem more ill-conditioned [111]. As PINNs are thoroughly based on empirical equations, an instability and non-convergence issues may arise during the training process [162].

7. ANN model development for FEM

ANNs are capable of surrogating high order models whilst acting as low order surrogates [163]. With the recent advancements in the field of ANNs, vast number of methodologies or algorithms and software tools are available and the integration in between ANNs and FEM can be performed in numerous ways. Also, it is worth noting that a single neural network model is designed for a particular structure and hence, it always becomes structure specific rather than becoming universally applicable [125]. Also, as ANNs are dealing with a dataset and are more statistical and conducting both forward and inverse analyses is possible. However, it is often questionable to decide on which ANN architecture is more appropriate for surrogating FEMs, the required size of the dataset to train the model, the most suitable activation function and loss function, the required computational capacity for the training process and so on. However, upon the comprehensive literature review that was conducted on this, it is clear that there is no universal way of estimating those parameters and those estimations are mostly dependent on the structure at interest, purpose of the model and previous experiences on same type of models [76]. It is more like a trial-and-error process. However, it is worth noting to identify the parameters of ANN models that are more common, popular and, which display excellent efficacy in surrogating FEMs leading to real-time simulations.

7.1. *Type and model architecture of the neural network*

It was observed that multi-layer perceptron, recurrent neural network, convolutional neural network, graph neural network and physics informed neural network are more popular in surrogating FE

Table 11
Applications of PINNs in surrogating FE modelling.

Author	Purpose	Size of the Data set	Data Preparation	Structure of the ML model	Activation Function	Model training	Inference time	Performance
Zhang et al. [171]	To simulate seismic effects on a 3-storey building	0.8/0.2 for training and validation. Testing – 760Training – 46	Non-linear FE modelling using RT-Frame 2D in MATLAB	3 deep LSTM networks (PhyLSTM^3)	Tan hfunction	Adam optimizer with Lr − 0.001, decay rate – 0.0001, epochs - 10,000	10^3 times faster than FEM	Over 0.9 correlation
Haghighat et al. [53]	To predict stress, strain and displacement variation of a square (linear)	10,000	Based on 2D elastic plain-strain analysis	Feedforward network with 5 hidden layers	Tan hfunction	Adam optimizer was used with batch size of 64, epochs – 10,000 and patience – 500.	Not mentioned	In a form of 10^{-5}
	To predict stress, strain and displacement variation of a perforated strip with a circular hole	2000	COMSOL software was used for plastic analysis	Feedforward network with 4 layers, each with 100 neurons				MSE is different for each parameter (range – 0.035 to 0.12)
Huzni et al. [64]	To simulate the reinforced concrete corrosion	10,000	2D FE analysis using Wolfram Mathematica 12	Not clearly mentioned	Not mentioned	Trained by checking the predicted data aligns with observed data whilst satisfying Laplace's equation for the problem.	88 ms	Standard deviation is less than 0.2
Rezaei et al. [135]	Solving PDEs in heterogeneous domains like linear elasticity and steady-state diffusion	5000 collocation points and 9261 collocation points for 3D extension	FE simulation considering the regular quadrilateral mesh	Feedforward neural network with 5 hidden layers and 40 neurons in each layer.	Tan hfunction	Trained by Adam optimizer for 100,000 epochs under the Lr = 0.001.	Not mentioned	4 % relative error for displacement and 3.7 % relative error for stress
Le-Duc et al. [86]	For parametric simulation of static problems −Planar 10 bar truss −Spatial 72 bar truss −Spring supported Euler-Bernoulli beam −Squared Reissner-mindlin plate	Two datasets – clean and noisy 9000 for training and 1000 for testing 800 for training and 200 for testing	Geometric and parametric features were inputs and nodal displacements considered via high fidelity FEM simulators	4-layer model with two hidden layers with 100 nodes on each	Tan h function	Adam optimizer is used with maximum epoch of 500 for the plate and 2000 for the rest. Lr = 0.001, exponential decay rates = 0.9 and 0.999	Not mentioned	MAE for interpolation case – 1.33E-4 3.04E-2 9.04E-3
Bai et al. [5]	Stress and deflection prediction of a pure bending of beams	125 points (domain) + 60 points (boundary)	Uniformly distributed sample points, plane strain assumption, Lamé constants used for constitutive model	5-layer feedforward neural network (FNN) with 10 neurons per hidden layer	Tan h function	Using Limited-memory Broyden–Fletcher–Goldfarb–Shanno with Box constraints (L-BFGS-B)	Not mentioned	~2–3 % error in displacements, 5–6 % in stress predictions.
	Stress and deflection prediction of a perforated strip plate	441 in-domain sample points + 120 boundary points	Uniformly distributed sample points, plane stress assumption, Abaqus used for FEM reference solutions	8-layer feedforward neural network (FNN) with 20 neurons per hidden layer				~1–3 % error for displacements, ~5% for stress, some deviations in boundary stress predictions
	Stress and deflection prediction of a cube with a hollow sphere	8726 uniformly distributed sample points	Uniform and random distributions considered, first-order Gaussian quadrature for integration, FEM for reference	7-layer feedforward neural network (FNN) with 20 neurons per hidden layer				Error in displacement was around ~ 2–4 %, and ~ 5–7 % for stress predictions.
Bastek and Kochmann [9]	Stress and deflection prediction of a hyperbolic paraboloid shell	2,048 collocation points	Sampled uniformly using Sobol sequence; gravity loading	3 hidden layers with 50 neurons per layer	GELU (Gaussian Error Linear Unit) function	Trained using L-BFGS optimizer; ~100 epochs for optimal convergence	Fast computation times	Average relative L_2-error ~ 5 %
	Vertical deflection prediction of Scordelis-Lo roof benchmark	65,536 collocation points	Uniform sampling with Sobol sequence; boundary conditions applied for clamped edges			Trained using L-BFGS optimizer for 1,000 epochs		Displacement u_3 − error ~ 1.7 % compared to FEM

(*continued on next page*)

Table 11 (*continued*)

Author	Purpose	Size of the Data set	Data Preparation	Structure of the ML model	Activation Function	Model training	Inference time	Performance
	Predicting displacement of a fully clamped hemisphere	78,400 collocation points	Gaussian load applied at center; clamped boundary encoded in trial function			Trained for ~ 100 epochs using weak form		Displacement L_2 −error ~ 3–4 %, stress error slightly higher
Vahab et al. [160]	Predicting displacement and stress fields in pile-soil interaction	6000 collocation points (3000 per domain)	Sampling grid with clustered points near the boundaries and interfaces; non-uniform sampling grid	4 hidden layers with 20 neurons per layer	Tan h function	Trained using Adam optimizer with a learning rate of 0.003 and Neural Tangent Kernel (NTK) adaptive weighting; converges to relative error 10^{-6} within ~ 1000 epochs	Not mentioned	Normalized displacement error ~ 10^{-2}
	Predicting deformation and stress in sheet pile wall		Plane strain assumption, simplified for long structural members; 3D sampling grid for boundary conditions			Adam optimizer used for training, converging to relative error of 10^{-5} within ~ 500 epochs		
	Identifying material parameters of layered soils	9000 points total (including 2000 input points)	Data acquired via synthetic strain profile along pile, with a 1D profile extracted from FEM solution	6 distinct networks for each displacement component		Trained using Adam optimizer with additional contact constraints; convergence to relative error norm 10^{-4} within ~ 1000 epochs		Less than 5 %
Chen et al. [21]	Predicting large deflection in cantilever beams	200 collocation points	Fixed and free boundary conditions	3 hidden layers with 10 neurons per layer	Tan h function	Trained with Adam optimizer; pre-training with 10,000 epochs	Not mentioned	Results closely matched with the FEM results
	Predicting load–displacement curve of imperfect columns		Pin-pinned and pin-fixed boundary conditions					
	Predicting large deflection in cantilever beams under distributed loads		Fixed and free boundary conditions					
	Predicting load–displacement curves in columns under combined loading		Combined axial and bending loading					
Wang et al. [162]	To simulate a tension problem of a plate with a hole in the middle	10,000 iterations	2D linear elastic analysis using Abaqus	Feedforward network with 7 hidden layers	Tan h function	Adam optimizer with Lr – 0.005, decay – 0.1 after 5000 iterations	–	The lowest MSE was 0.00033

Table 12
Comparison of different neural network architecture in surrogating FE modelling.

Type of the Neural Network	Advantages	Disadvantages
Multi-Layer Perceptron (MLP)	• Simple straightforward architecture • High generalization capacity • Effectively learns with backpropagation algorithm • Can be used for both linear/non-linear problems	• Inability to represent time history analysis problems. • Susceptible to overfitting without adequate regularization • Does not consider underlying physics • Inability to capture spatial relationships
Recurrent Neural Network (RNN)	• Ability to surrogate time history analysis problems • Independency from mesh resolution and complexity • Can be used for both linear/nonlinear problems	• Prolonged training time due to vanishing gradient issues • Overfitting due to high generalizability • Can be computationally intensive in real-time FE modelling
Convolutional Neural Network (CNN)	• High feature detection ability • Handling multi-dimensional data • Ability of showing spatial relationships in a FE mesh	• Requirement of fixed input tensor size • Inability of handling variations in rotations and scale • Limited extrapolation capacity
Graph Neural Network (GNN)	• Irregularities and discontinuities of FE mesh can be considered. • Represents connections and dependencies of graph nodes • Message passing ability in between nodes	• Memory demanding nature and lack of generalizability • Limited extrapolation capacity • Over smoothing problems may rise due to extensive layering
Physics Informed Neural Network (PINN)	• Represents underlying physics of the system • Less number of training data are required • High sensitivity to both interpolation and extrapolation processes	• Requirement of large training times • Instability and non-convergence issues may arise in the training process. • Limited capacity of surrogating complex FE meshes

modelling whilst radial basis functions and autoencoders are also in use depending on the intended purpose of the analysis process. In some scenarios, a combination of two or few types of neural network architectures can be identified to enhance the functionality of the surrogate model. For instance, to introduce the time factor to the mesh based FE model, a RNN can be introduced even if the main analysis process is based on CNNs [47]. Table 12 compares the strengths and weaknesses of each neural network architecture when surrogating FE modelling.

It is highly notable that neural network architectures, with the exception of PINNs, have limited ability to extrapolate beyond the range of data that were trained on. Hence, it is highly significant to have a priorly simulation of the FE model in order to identify the failure capacity of a model and to define the input and output data whilst representing that exact range. However, PINNs are free from this issue as they are additionally trained considering the underlying physics of a system. MLPs can be identified as the most simple and straightforward architecture that can do the surrogation process with minimal effort. However, other models require a significant effort to conduct a pre analysis process to transform the FE model into the necessary format for model training. However, for any type of problem that is incorporated with the time factor, the application of recurrent neural network is required unless considering inputs in a two consecutive time frames for other NN architectures [47]. Both CNNs and GNNs can be identified as the most effective architectures for surrogation of FE modelling due to their ability of handling spatial data. However, GNNs outperform the CNNs due to their inherent ability of dealing with irregularities and discontinuities in FE meshes. The most appropriate neural network architecture is always varied depending on the intended purpose. However, GNNs can be identified as the preferred neural network architecture for surrogating a complex FE mesh when comparing with other types.

It was identified that in the surrogation process of finite element analysis, the tested neural networks always have limited number of intended purposes such as predicting structural behaviour, determining structural/material properties, determining structural stability or parametric simulations. Fig. 11 represents the identified literature and categorizes them with the intended purpose of each case. It basically maps out the connections in between the neural network model with their targeted predictive tasks. The width of each connection indicates the proportion of studies for each combination. The majority of the surrogate modelling efforts of FEM have focused on predicting stress and displacements (either nodal or maximum displacement) of the interested structure. Whilst that type of prediction is significant in terms of understanding structural behaviour, it is well justified the reason for making them essential targets of surrogate modelling as they directly influence the structural integrity. Stress or strain or displacement estimations are the basis of structural analysis to estimate the safety, durability and overall structural behaviour under various conditions. Apart from that, the prediction of failure, factor of safety (FOS) and material property estimations have also been less represented, indicating the potential opportunities for future research. Further, it can be clearly seen that MLPs demonstrate significant potential in predicting displacement as they are well suited for approximating the non-linear relationships between loads and structural deformations. RNNs are mostly equally contributed for both displacement and stress predictions whilst CNNs and GNNs show a widespread application of predicting stress. PINNs have been used for a wide variety of purposes as they utilize physical laws for the training process. Following the detailed analysis presented in Fig. 11, Fig. 12 represents broader application domains of these neural networks, by categorizing them as per the analysis process or analysis type. It is to investigate such a scenario as it offers a holistic perspective on the role of neural networks in addressing multifaceted challenges and future research areas.

From Fig. 12, it is worth noting that the static structural behaviour analysis has been more commonly done in the previous literature where a significant number of literatures has solely focused on forward analysis of FE modelling. Inverse analysis has achieved a less attention with only MLPs and PINNs demonstrating a significant focus on simulating that. This limited engagement of ANNs suggest an opportunity for further exploration, where the strengths of these neural network types could be leveraged to advance inverse modelling techniques in structural analysis. It is further clear that MLPs, RNNs and PINNs are being used for diverse tasks due to their flexibility and simple model architecture whilst a notable contribution of CNNs and GNNs can be identified on either static structural behaviour or material property estimations as they capitalise on their strengths in handling spatially structured data. However, a small fraction of CNNs and GNNs has paid attention on dynamic analysis of structures harnessing the potential of these networks to better understand the dynamics of structures. Further, parametric simulations were mostly done using MLPs, RNNs and PINNs as their flexibility to handle varying numerical input data unlike the CNNs and GNNs which are better suited for spatial and relational data.

7.2. The dataset

The quality and pre-processing of the data are two critical aspects in creating ANN models to surrogate FEM. High-quality data ensures the reliability of forecasts and preciseness of relationships of the model whilst the poor-quality data often leads to erroneous or biased models jeopardizing the reliability of FEM simulations. Further, the dataset should be a complete representation of the problem including all the necessary aspects by using a validated FE model. Even though the

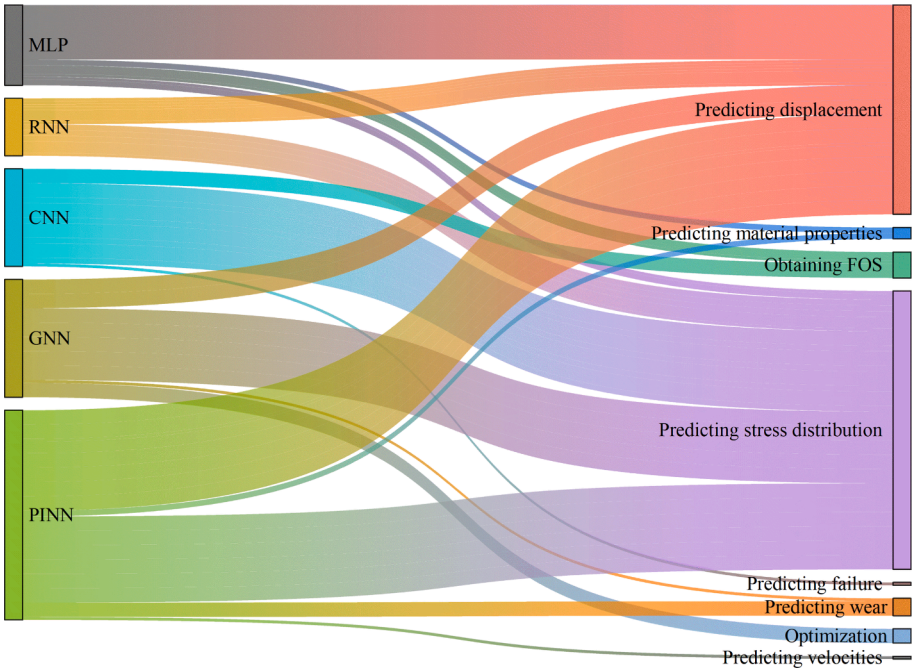


Fig. 11. Relationships in between the type of the neural network and the intended purpose.

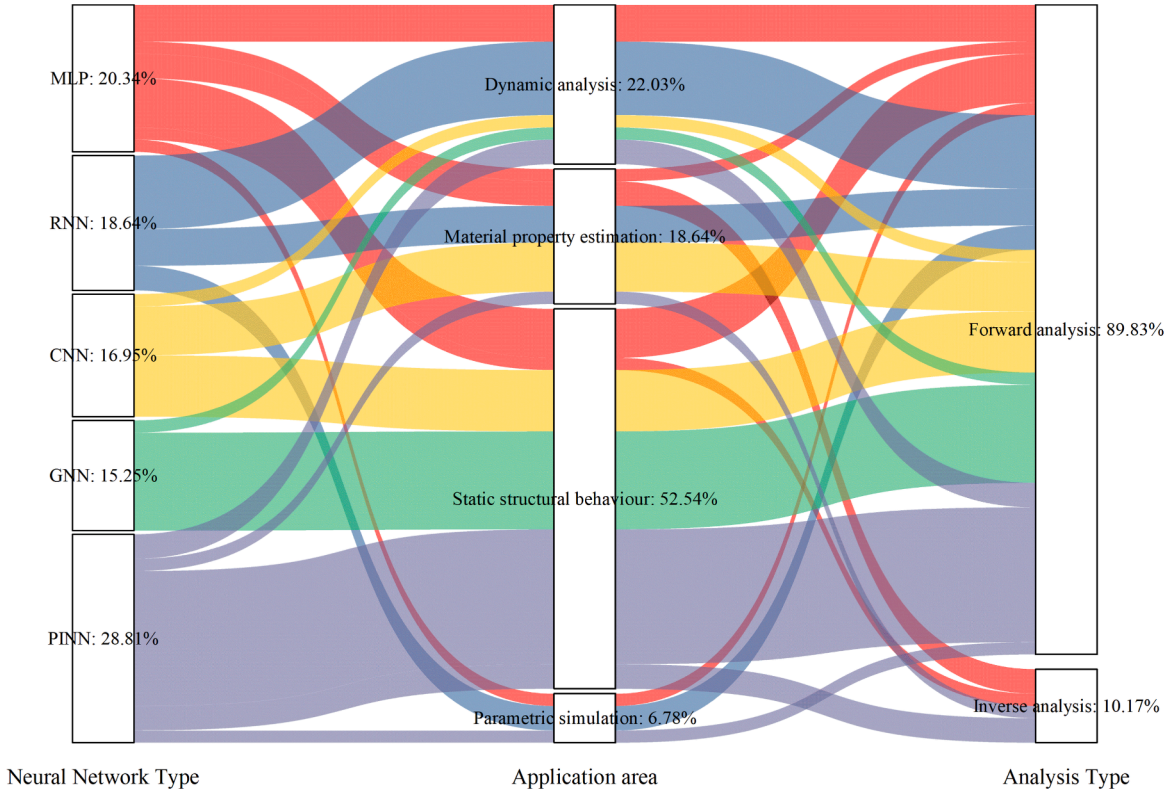


Fig. 12. Applications of different types of Neural Networks in different areas.

dataset is prepared with a highly accurate FE model, the data pre-processing is highly significant as well. It is recommended to check the dataset for outliers or duplicates and scale the data to enhance its predicting accuracy. Different scaling methods such as Min-max or Z-score can be used. Also, if it is needed, the data should be transformed as well for a suitable format depending on the ANN model.

Also, it is always questionable to determine the required size of the dataset to successfully train the model to surrogate the FE analysis procedure. It is highly impossible to get an idea about the optimum amount of data for this purpose from previous studies. Except PINNs, other studied models require a higher amount of data to be trained when compared with PINNs. However, with the assistance of physical loss functions, PINNs can easily reduce the required amount of data for a successful training. Overall, if more the data are available, the model

tends to be more accurate if the other training parameters are controlled during the training process to avoid overfitting. The rule of thumb for the size of the dataset to be at least 10 times of the number of parameters of the model. For data generation, commercially available FE software, a developed script on FE analysis or even experimental data can be used. Also, another important consideration is that the FE model should be validated against the experimentally measured data to ensure a high-quality dataset that accurately represents the actual structure. Otherwise, even if the ANN-based surrogate model is trained to replicate FE-based data, it may fail to effectively simulate the actual structure [67]. The data generation procedure is time consuming and computationally intensive. However, the later benefits outweigh the computational intensity of the data generation procedure.

It was identified that the dataset was split into subsections that include training, testing and validation datasets. The training set is used for training the model whilst the testing set is to evaluate the performance of the model, and the validation set fine-tunes the model by adjusting hyperparameters. However, the choice of proposing a validation dataset is optional. The most common train/test split was found to be 80/20. It means 80 % of data for training and 20 % of data for testing. In addition to that, a validation dataset was formed by further splitting the testing data. Some studies have not considered a validation dataset and conducted the training by only considering the training data. However, authors recommend the use of a validation dataset to further enhance the accuracy and the quality of the trained neural network model. In addition to the 80/20 split, 70/30, 60/40, 90/10, 85/15, 95/5, 75/25 and 94/6 were identified as possible training to testing data split. Cross validation has also been used for training and testing split to further enhance the reliability of the model whilst ensuring the participation of each data point for both training and testing. This process is further known as k-fold cross validation as the dataset is divided into k number of subsets, trained and tested for k times using different subsets at each time. Fig. 13 depicts certain ways of improving the quality of the dataset for the surrogation process as per the analysis of previous literature.

7.3. Activation function and loss function

An activation function is being used to determine the output of a neural network and some of the commonly used activation functions have been summarized in Table 13, under two folds: linear and non-linear activation functions (In Table 13, $\boldsymbol{a}$ refers to a constant whilst $\boldsymbol{c}$ refers to the center and $\boldsymbol{\sigma}$ refers to the standard deviation of the data set). Non-linear activation functions are used to add a non-linearity to the neural network and consequently, the model tends to learn more complex representation of data [105]. ReLU function is most commonly used for CNNs [130], and it would be an excellent initial choice when it is unsure on which activation function to employ. However, the dying ReLU problem may occur in exceptional cases. In reality, if a neuron becomes negative and remains same regardless of inputs, it is considered dead and does not contribute to the learning process [103]. To address this, the Leaky ReLU has been introduced. The application of sigmoid function is also popular for neural networks. However, its gradients may become very small or vanish during back propagation due to any saturations of large positive and negative values. Also, the sigmoid function must be zero-centered to avoid zigzagging while training [12]. Hyperbolic tangent or tanh function is recommended to use in recurrent neural networks and similar to the sigmoid function, the vanishing gradient issue will be there [88]as well. Likewise, activation functions are inherent with various characteristics and hence, it is required to select the best activation function depending on the type of the neural network and also depending on the purpose [12].

The Leaky ReLU and ReLU activation functions are more commonly used for MLPs, RNNs, CNNs and GNNs whilst for PINNs, the tan h (hyperbolic tangent) function was commonly used due to its continuous nature [53,86]. Apart from that, the sigmoid function was also identified as a suitable activation function for surrogating FEMs. In some scenarios, different types of activation functions were used for different layers of the model [24]and some cases a combination of activation functions were also used with the purpose of maximizing the accuracy of the model [170]. However, as per the review, it is clear that the ReLU and Leaky ReLU are more popular than other activation functions. ReLU activation function can be recommended as an effective starting point for a neural network model due to its success in assisting for the

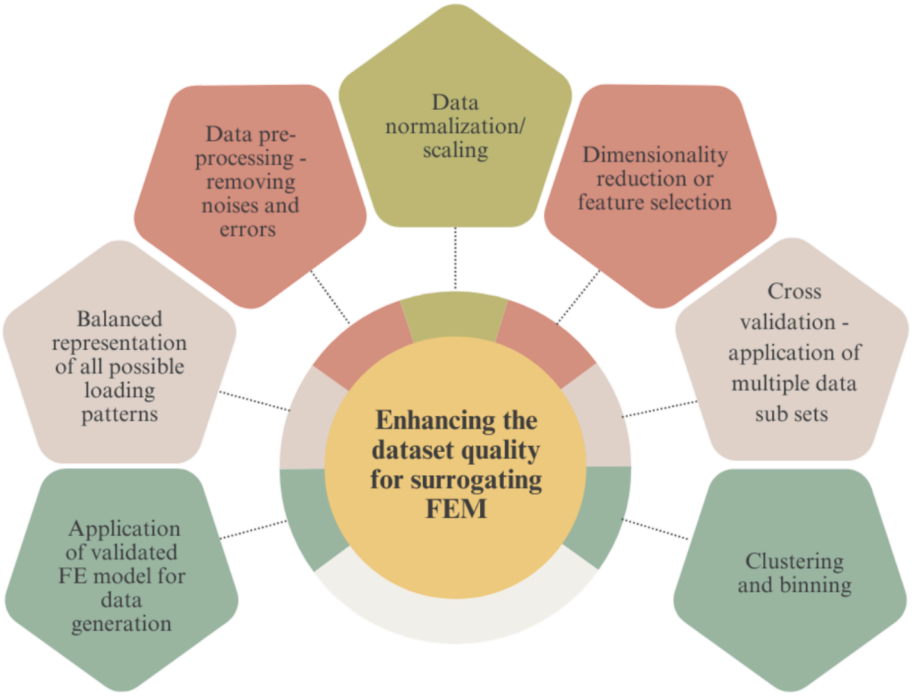
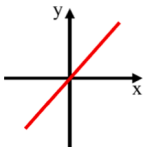
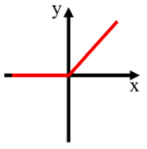
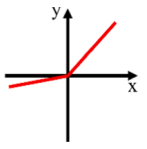
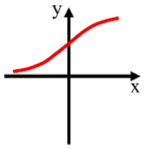
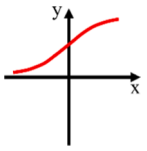
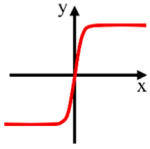
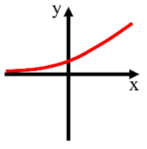
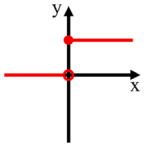
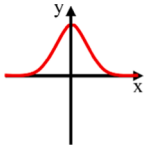


Fig. 13. Different ways of enhancing the data quality for surrogating FEM.

Table 13
Commonly used activation functions.

Function	Equation	Graph	Range
Linear activation functions			
Linear	$f(x) = x$		$(-\infty, +\infty)$
Non-linear activation functions			
Rectified Linear Unit (ReLU)	$f(x) = \max(0, x)$		$[0, \infty)$
Leaky ReLU	$f(x) = \max(ax, x)$		$x > 0 - [0, \infty) x < 0 - (-\infty, \infty)$
Sigmoid or logistic	$f(x) = \frac{1}{1+e^{-x}}$		$(0, 1)$
Log-sigmoid	$f(x) = ln\left(\frac{1}{1+e^{-x}}\right)$		$(0, 1)$
tanh or hyperbolic tangent	$f(x) = \tanh(x)$		$(-1, 1)$
Soft plus	$f(x) = \ln(1 + e^x)$		$(0, \infty)$
Binary step	$f(x) = 1; x \geq 0 f(x) = 0; x < 0$		$\{0, 1\}$
Radial Basis	$f(x) = e^{\frac{-(x-c)^2}{2\times\sigma^2}}$		$[0, 1]$

surrogation process of FE models. However, if a neuron becomes zero in the training process, it will be avoided from the training process by ReLU activation functions. The Leaky ReLU function has been introduced to avoid such an issue and it has also been a huge success. However, activation functions are also affecting the performance of a neural network model and it is really crucial to select the most appropriate activation function for the process [12]. Further, it is noticeable that PINNs often use the hyperbolic tangent or tanh function as the activation function due to its ability to providing smooth and continuous gradients whilst symmetrically handling both positive and negative values for better approximations of physical systems. Sigmoid function also provides smooth and continuous gradients. However, it is less symmetric when compared with tan h as it is within the range of [0,1] and that might slower the convergence when dealing with physics-based functions.

Loss functions also play a crucial role in the training process of a neural network model. Mean squared error (MSE), mean absolute error (MAE), relative mean squared error (RMSE), mean relative error (MRE) and mean absolute percentage error (MAPE) are most commonly identified loss functions in the studied literature. Always it is required to consider a squared error or an absolute error to overcome the issue of positive and negative errors cancelling each other out. Only PINNs follow a different set of functions that reflects the underlying theory. Bai et al. [5] categorized the loss functions in PINNs into two major categories: collocation loss functions and energy-based loss functions. The collocation loss is calculating the MSE of residuals of sample points and summing them up. Consequently, the degree in which the response of the model address the physical laws will be identified [53]. However, the residuals may be in different scales as both governing equations and boundary conditions are included in physical equations of a system. Hence, some residuals may numerically ignored due to their minuteness and the challenge of accurately summing them up, especially when they are in different orders of magnitude [5]. The energy-based loss function accounts the theory of having minimum energy at the equilibrium state [141]. However, only the displacement prediction was successful with energy based loss functions and not for stress fields as they are unable to satisfy the equilibrium and boundary conditions together [35]. A novel loss function, least squared weighted residual (LSWR) has been introduced whilst addressing the limitations of both collocation and energy based loss functions [5].

7.4. Training of NNs

Training neural networks (NNs) for surrogate modelling of FEM presents unique challenges and opportunities that extend beyond generic training practices. While widely available libraries such as TensorFlow, Keras, and PyTorch provide the foundational tools for implementing ANN models, the specific requirements of FEM-based applications demand tailored approaches. For model training, all the functions required for the loss function, activation function, optimizer function and training process are pre-defined in the libraries. Hence, it is only required to decide the functions which are required for the model implementation and training of the model and simply call those functions to conduct the process. However, determining the model parameters, specifically for the surrogation, is still challenging and mostly structure specific. When selecting hyperparameters for training, structural applications often require fine-tuning of key parameters such as learning rates, batch sizes, and patience settings for early stopping. These parameters directly impact the convergence of the model and its ability to generalize to new data. Additionally, optimizer choices play a significant role in training efficiency; Adam optimizer remains a popular choice in this domain due to its adaptive learning rates and momentum-based updates, which are particularly effective for non-convex optimization problems common in neural networks for FEM tasks [81].

Further, one of the critical considerations that are identified in this domain in model training is the incorporation of domain-specific constraints into the training process. PINNs, for example, embed governing equations like equilibrium or compatibility conditions directly into the loss function, ensuring that the trained model respects fundamental physical principles. This approach is especially beneficial for FEM applications where accuracy and physical consistency are paramount. Training efficiency is another major aspect, given the computational

intensity of FEM simulations. Leveraging graphical processing units (GPUs) instead of central processing units (CPUs) for parallel processing can significantly accelerate training, making it feasible to handle large-scale datasets. Emerging frameworks, such as GNN-based architectures for mesh-based data, further enhance the applicability of NNs for FEM problems by directly processing spatially varied data in the training process. Also, evaluating model performance required domain-specific metrics is another key consideration. While traditional metrics like mean squared error (MSE) are useful, FEM-specific evaluations such as accuracy of stress or strain predictions under variable loads provide more actionable insights. These metrics ensure that the trained model meets the precision requirements necessary for real-world applications. Ultimately, it is clear that while general NN training principles provide a foundation, the training of NNs for FEM applications requires a tailored approach that considers domain-specific challenges, integrates physical constraints, and leverages advanced computational tools. Such approaches pave the way for creating reliable surrogate models capable of real-time predictions in structural analysis.

7.5. Real-time simulation with ANNs

Conducting real-time simulations is a critical requirement in various engineering applications, including structural health monitoring [177,179], digital twins [67], and emergency response systems [67]. While FEM is a robust and accurate tool for structural analysis, its computational intensity makes it impractical for real-time use when employed as the sole method [167]. In contrast, ANNs provide a computationally efficient alternative for real-time simulation by serving as surrogate models for FEM. However, achieving real-time performance with FEM using ANNs involves addressing several key considerations and following a defined pathway.

The term 'real-time' in the context of ANN-based methods refers specifically to the inference stage, when a pre-trained model provides instantaneous predictions of structural responses based on input parameters. This capability is enabled by the substantial computational investment made during the training phase. Once trained, ANNs can bypass the iterative solution processes inherent in FEM [100], delivering predictions within milliseconds [98,59]. This makes them particularly valuable for applications requiring rapid decision-making, such as monitoring critical infrastructure during seismic events or extreme weather conditions [67]. As mentioned in section 5.1, authors identified three main approaches that are commonly used for reducing computational time ANNs. The second approach offers a more comprehensive method for surrogating FEM and is often more successful, as it focuses solely on the required inputs and outputs. Therefore, this approach has been emphasized in this section to achieve near real-time computations. However, the most crucial factor in this context is that the real-time performance of ANN-based surrogates for FEM is not determined during the training phase. Real-time analysis is only achievable after the model has been fully trained. Fig. 14 presents the step-by-step methodology of achieving near real-time surrogate model for FEM via ANNs.

As depicted in Fig. 14, the step-by-step methodology can be presented in three stages: (I) Data preparation, (II) Training the ANN and (III) Real-time data input. Upon considering the structure of interest, the FE model of that should be first developed. Subsequently, it must be validated by analyzing the structural response under known loads or through established methods such as Frequency Domain Decomposition (FDD) method [15,118]. The required input and output values can then be determined based on the specific purpose, and a comprehensive dataset can be generated by considering various scenarios. For example, in the case of a parametric analysis, parameters such as geometry or material properties can be varied within a defined range, and the corresponding output values can be calculated. This process leads to the creation of a comprehensive data set. In the second step, once the dataset is prepared, the appropriate ANN architecture must be selected. For example, if the results need to be presented in a mesh-based format, GNNs would be a suitable choice. Alternatively, for time-dependent behavior, RNNs would be more appropriate. Next, the ANN model can be defined by selecting optimal hyperparameters and trained until it achieves sufficient accuracy. Once trained, the model can be tested and further refined if necessary to improve performance. In step three, the parameters can be input into the trained model at an updating frequency appropriate for the problem. Once the model processes the input data, it generates outputs under near real-time conditions. Finally, the results can be visualized, and further analysis can be performed, if necessary, in real time. Hence, the real-time analysis occurs exclusively during the third step. The first and second steps serve as pre-analysis phases, which are essential for creating the framework necessary to enable the real-time process. Also, this process is not as straightforward as it may seem, as it involves several challenges and limitations as mentioned in section 7.6. Nonetheless, for applications requiring immediate decision-making, the value of such ANN-based surrogates is undeniable.

7.6. Limitations and challenges in ANN-Based surrogates for FEM

Through the comprehensive review conducted on ANN-based surrogates for FEM, several limitations and challenges have been identified. This section provides an in-depth discussion of these issues as a summary, highlighting the key barriers to their effective implementation, and potential areas for further research.

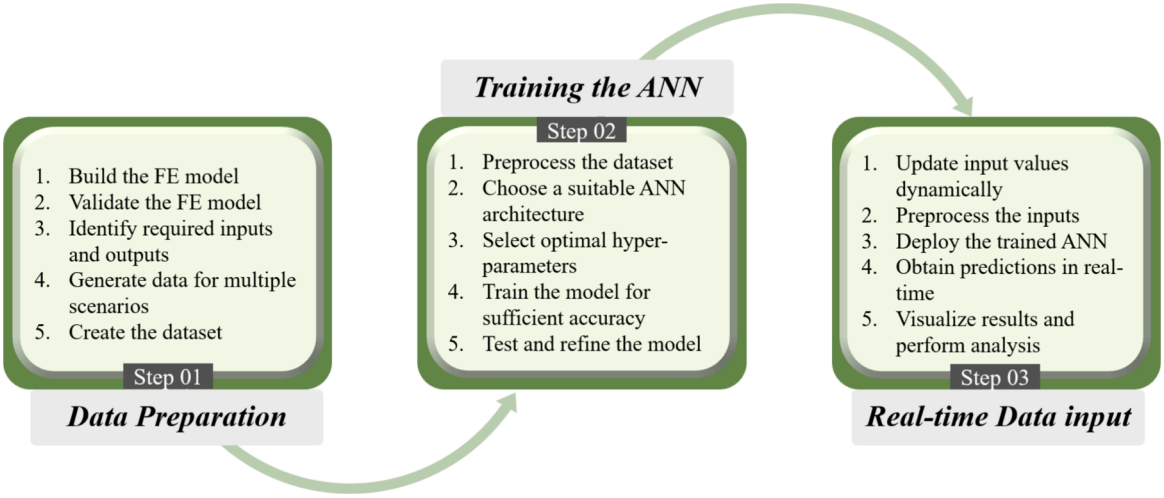


Fig. 14. Step-by-step methodology of achieving real-time analysis.

1. Data requirement

ANNs usually rely on extensive, high-quality datasets for effective training. When the dataset comprehensively captures the behaviour of the system, the model can be trained to accurately represent the required outputs. In the context of FEM, generating such datasets demands significant computational effort, particularly for large-scale or highly detailed models. Additionally, since the dataset is derived from FE analysis, the FE model must be validated to ensure sufficient accuracy in representing the real structure. This validation is crucial and can be achieved through techniques such as FE model updating. Also, the literature review indicates that there is no specific limit on the size of the dataset; in fact; the availability of more data generally leads to more accurate results. However, the most critical factor is that the dataset must fairly represent the FE model and consist of high-quality data.

2. Generalization

Another significant challenge faced by ANN-based surrogate FEMs is the generalization. Generalization refers to the ability of ANNs to handle unseen scenarios, which becomes problematic when the training data does not adequately cover the range of possible conditions. The review of literature reveals that researchers have attempted to address this challenge by developing datasets that encompass a wide range of loading patterns, including ultimate scenarios. Trained models are even capable of generating accurate predictions avoiding slight deviations of the input values in the covered range. However, it has been identified that the extrapolation capacity of traditional ANNs is limited. Hence, if the input values fall outside of the range of scenarios that are covered in the in the dataset, the trained model may struggle to predict accurate results. However, PINNs offer a promising solution in such cases, as their loss function incorporates physical relationships, enhancing their ability to generalize. Consequently, in situations where the ultimate loading scenarios is not well understood, incorporating a carefully designed loss function that reflects physical principles is crucial for addressing generalization issues. Additionally, ANN-based surrogate FEMs are structure-specific, requiring the development of a new surrogate model from scratch for each individual structure. It will be really challenging to implement a trained ANN that works on any structure.

3. Computational intensity in model training

FE models often incorporate a large number of parameters, which significantly increases the effort required for training to achieve a well-trained model. Additionally, the high number of parameters necessitates substantial memory capacity to store the training data. During training, the model must handle predictions involving a large volume of output parameters, which demands considerable processing power. Consequently, most studies have opted for GPUs over CPUs for model training, as GPUs provide the computational efficiency needed for such intensive tasks.

Even though the model inference becomes real-time, the data preparation and all other steps are not real-time.

4. Lack of interpretability

FE modelling relies on a vast number of physical principles, where the structural behaviour is fully explained by one or more well-defined physical relationships. In contrast, ANNs operate as black-box models, meaning the internal processes during training are not easily interpretable. This black-box nature poses challenges in identifying errors and validating predictions, particularly in critical engineering applications where accuracy is paramount. ANNs essentially approximate the relationships between inputs and outputs. However, if the training data is incomplete, biased, or does not adequately represent the structure, the predictions may lack accuracy. As a result, ANN-based surrogates may not achieve the same precision as direct FEM simulations, even though they can produce results much faster. This limitation makes ANNs less suitable for applications requiring highly precise computations but valuable for scenarios where approximate insights into structural behaviour are sufficient and computational efficiency is a priority.

5. Challenges of making the process real-time

Although ANNs significantly reduce computational time compared to FEM once trained, the training process itself can be time-intensive, as it requires generalizations across all parameters. Consequently, only the computational phase can achieve near real-time performance, not the entire process. For scenarios requiring instant results, pre-trained ANN models are ideal. However, achieving near real-time performance for much larger structures remains a significant challenge, even with ANNs, due to the complexity of the models. Despite this, ANN-based surrogates still consume less computational time than FEM for predictions. It is also worth noting that real-time applications rely heavily on hardware acceleration, which may not always be accessible for practical for all users.

6. Replication of FE model complexity

While ANNs offer a powerful alternative for FEM tasks, they face limitations in fully capturing the intricate behaviours and multi-physics interactions modelled by FEM. ANNs generate outputs based on a given set of inputs, rather than comprehensively accounting for the underlying complexity of FEM. Their integration into FEM workflows can address computational bottlenecks, but replicating the full complexity of FE models will require advancements in data representation, training methodologies and hybrid approaches such as PINNs. Although PINNs embed physical laws into the loss function and can account for some aspects of model complexity, they are not yet highly effective for surrogating highly complex FEM scenarios. Therefore, ANNs are best recognized as complementary tools rather than direct replacements for FEM.

As a summary of the above discussion, Table 14 summarizes the key trade-offs between ANNs and FEM.

Table 14
Key trade-offs between ANNs and FEM.

Aspect	ANNs	FEM
Computational speed	Fast once trained	Computationally intensive
Interpretability	Limited	High
Precision	Lower for complex scenarios	High even for novel cases
Generalization	Relies on data quality	Robust across varied cases
Real-time applicability	Ideal for real-time outputs	Challenging for real-time outputs
Replication of model complexity	May struggle to replicate highly complex scenarios	High accuracy even in complex scenarios

Table 15
Comparison of alternative ML models for integration with FEM.

ML Method	Principle	Advantages	Challenges	Applications	References
GANs	Generative modelling using adversarial training (generator vs discriminator)	– Captures complex distributions – Augments datasets – Creating high-quality simulations	– Training instability – Mode collapse – High computational cost	Generating statistically equivalent 3D microstructures as high-fidelity inputs for multiscale FEM simulations to analyse material behaviour.	Murgas et al. [120]
				Generating synthetic structural datasets to enhance FEM-based reliability assessments of steel portal frames.	Ljubinković et al. [99]
RL	Trail-and-error learning with reward-based optimization	– Effective in optimization and decision-making tasks	– Long training times – Complex reward function design	To automate the generation of high-quality meshes for FEM	Pan et al. [128]
				To identify critical elements and plan demolition in frame structures	Zhu et al. [174]
TL	Leveraging pre-trained models for related tasks to reduce training requirements	– Reduces training time and data requirement	– Domain adaption issues – Limited pre-trained models for FEM	For structural condition identification.	Bao et al. [8]
				To select features before applying transfer learning for damage detection.	de Almeida et al. [26]
Diffusion models	Iterative denoising process to generate high-quality output from random noise	– Produces high quality data with fine details	– Computationally intensive – Limited engineering applications	Structural topology optimization	Mazé and Ahmed [110]
Transformers	Probabilistic inference to model uncertainty in structural behavior	– Explicit uncertainty quantification – Robust reliability insights	– Computationally expensive – Sensitive to prior assumptions	Foundation models for Structural Health Monitoring (SHM)	Benfenati et al. [10]
Bayesian probabilistic models	Probabilistic inference to model uncertainty in structural behavior	– Explicit uncertainty quantification – Robust reliability analysis	– Computationally expensive – Sensitive to prior assumptions	Reliability analysis of a 2D truss, simply supported beam and complex subway station	Zhang et al. [173]
				Likelihood approximation and computational cost reduction in structural reliability analysis	Li et al. [92]
				Bayesian FE model updating and uncertainty quantification for damage detection	Li et al. [93]
Support vector machines (SVMs)	Maximizes the margin between data classes using hyperplanes in high-dimensional space	– Robust for small datasets – Effective for classification tasks	– Sensitive to hyper parameter tuning – Less effective for large datasets	To modify FEM models for reinforced concrete beams subjected to elevated temperatures	Liu et al. [97]
				To forecast the compressive strength and Young's modulus of lightweight aggregate concrete	Bonifácio et al. [14]
				Surrogating FEM for damage identification in dams to improve computational efficiency	Li et al. [94]
Ensemble models	Combines multiple weak learners (i.e. decision trees) to improve overall performance	– High accuracy – Robust to overfitting	– Computationally expensive – Less interpretable	Hybrid stacking ensemble method that integrates finite element analysis and field data to predict slope stability	Kardani et al. [75]
				A neural network ensemble to update finite element models.	He and Yin (2023)
Gradient Boosting	Sequentially improves weak models by minimizing residual errors	– High accuracy – Feature importance insights	– Sensitive to overfitting and hyperparameter tuning	to predict uplift capacity factors for suction caissons in clay	Tran et al. [157]
				for predicting strain in Near-Surface Mounted Fiber-Reinforced Polymer (NSM-FRP) strengthened reinforced concrete beams	Khatir et al. [78]
				to estimate diaphragm wall deflections caused by excavation	Liang et al. [95]
Random Forest	Constructs multiple decision trees and aggregates their results for predictions	– Handles large datasets – Reduces overfitting	– Less effective for highly complex relationships	For converting solutions derived from the First-order Shear Deformation Theory (FSDT) into solutions based on elasticity theory.	Garg et al. [41]
				for deterministic finite element model updating	Kamali et al. [73]
				Predicting displacement of a truss structure	Jayasinghe et al. [68]
Clustering algorithms	Groups data into clusters based on similarity or distance metrics	– Unsupervised learning – Effective for data exploration	– Sensitive to initial conditions and cluster shapes	To partition points within polyhedral grids	Antonietti et al. [2]
				In identifying and debugging errors within FEM simulations	Kannan [74]
				For the optimal design of large industrial steel structures	Minh et al. [114]
Gaussian process regression (GPR)	Provides probabilistic predictions with uncertainty quantification	– Explicit uncertainty quantification – Flexibility	– Computationally expensive for large datasets	For updating structural finite element models using dynamic measurement data	Moravej et al. [116]
				To update finite element models by assimilating point cloud data	Graves et al. [45]
				Gaussian Process models with inverse finite element methods to pre-extrapolate strain and estimate uncertainties	Poloni et al. [131]

8. Comparing alternative ML methods with ANNs for FEM surrogates

While the primary focus of this study is to investigate the applications of ANNs in FEM-related tasks, it is equally important to explore other machine learning models that have shown promise in addressing similar challenges. These models, including generative adversarial network (GANs), reinforcement learning (RL), transfer learning (TL), diffusion models, transformers, Bayesian probabilistic approaches etc., offer unique advantages and capabilities that could complement or expand upon ANN-based methodologies. However, despite analyzing their potential, it is also crucial to evaluate the advantages of adopting ANNs for FEMs in comparison to these emerging technologies.

Table 15 provides an overview of alternative ML methods that have shown potential for FEM-related applications. It highlights the underlying principles of each method, their specific applications in engineering, key advantages, and challenges. This summary serves as a foundation for understanding how these methods compare to ANNs and their potential roles in advancing FEM surrogates and structural reliability analyses.

As outlined in Table 15, several emerging machine learning technologies have the potential to synergize with FEM, significantly enhancing its computational efficiency. GANs are being used in the field of civil engineering for crack detections and damage detection using image-based data [104,145]. It is seldom to see that GANs are directly being used for surrogating FEMs with the purpose of accelerating computational time. However, it is common to see that GANs have been used as data augmentation tool for FE related tasks in the civil engineering domain [120,99]. Reinforcement Learning (RL) has been utilized in civil engineering for tasks requiring optimization and decision-making [174]through reward-based learning processes. RL models are particularly effective in automating complex processes such as mesh generation and structural decision-making [128]. Despite its advantages, RL faces challenges such as extended training times and the complexity of designing appropriate reward functions tailored to specific civil engineering tasks. TL also offers a promising pathway to accelerate FEM-related workflows, particularly in scenarios where data availability or computational resources are constrained and it is effective for structural condition identification and feature selection in damage detection, allowing researchers to bypass extensive training processes [8]. Diffusion models, known for their iterative denoising process, have been identified as ideal tools for structural topology optimization, which is an important areas in civil engineering and also, it has been identified that conditional diffusion models outperform GANs in structural topology optimization [110]. Transformers, with their probabilistic inference capabilities, have shown promise in modelling uncertainty in structural behaviour, offering explicit quantification and robust reliability insights. These models are particularly suited for Structural Health Monitoring (SHM), as demonstrated by [10], who applied foundation models with Transformer architectures to enhance anomaly detection and traffic load estimation. Despite their strengths, including superior interpretability and accuracy, Transformers face challenges such as high computational demands and sensitivity to prior assumptions. Nevertheless, their potential to revolutionize FEM-related analyses in civil engineering makes them a compelling area for further exploration.

Bayesian probabilistic models are powerful tools for incorporating uncertainty into structural behaviour analysis, providing explicit quantification and robust reliability insights. These models excel in applications such as damage detection and reliability assessment, as seen in Li et al. [93] where Bayesian updating was employed for finite element models to detect structural damage effectively. Similarly, Zhang et al. [173] applied Bayesian methods to analyse the reliability of diverse structures, from 2D trusses to complex subway stations. While these models offer unparalleled precision, their computational expense and sensitivity to prior assumptions pose challenges. Efforts like those by Li et al. [92] aim to address these limitations by improving likelihood approximation and reducing computational costs, broadening their utility in civil engineering.

Support Vector Machines (SVMs), ensemble models, and gradient boosting have found diverse applications in civil engineering, particularly in enhancing FEM-related analyses. SVMs, known for their robustness with small datasets, have been used to surrogate FEM for tasks like damage identification in dams [94]and modifying FEM models for concrete beams under elevated temperatures [97]. Ensemble models, which combine multiple weak learners to improve accuracy, have been applied in slope stability prediction [75]and finite element model updating [57], offering robustness but with higher computational demands. Gradient boosting, recognized for sequential improvement of weak models, has been effective in predicting uplift capacity factors for suction caissons [157], strain in reinforced beams [78], and diaphragm wall deflections [95] though it remains sensitive to hyperparameter tuning [175]. Also, Random Forest, clustering algorithms, and Gaussian Process Regression (GPR) are widely used for FEM-related tasks in civil engineering, each offering unique advantages. Random Forest, with its ability to handle large datasets and reduce overfitting, has been applied for predicting truss displacements [68], updating deterministic FEMs [73], and converting FSDT-based solutions to elasticity theory [41]. Clustering algorithms excel in data exploration and unsupervised learning, aiding in error identification in FEM simulations [74], grid partitioning [2], and optimizing industrial steel structures [114]. Meanwhile, GPR offers probabilistic predictions with uncertainty quantification, making it ideal for FEM updates using point cloud data [45], dynamic measurements [116], and strain extrapolation [131]. Despite their strengths, these methods face challenges such as computational demands and sensitivity to initial conditions, which shape their application scope.

As per the conducted literature review, it is clear that ANNs remain at the forefront of real-time FE modelling due to their versatility, ability to capture complex non-linear relationships, and capacity to handle high-dimensional data. These attributes make ANNs particularly effective for applications such as structural health monitoring and digital twins. However, advancements in other models, such as GANs, RL, TL, Diffusion Models, Transformers, and Bayesian Probabilistic Models, have also shown potential in FEM-related tasks. Yet, these alternatives have limitations, and they are in early stages of development compared to the mature and flexible nature of ANNs. For instance, even though GANs excel in generating high-quality synthetic data and have been applied in image-based tasks like crack detection and data augmentation, they are computationally expensive to train and often lack interpretability compared to ANNs. Also, RL is effective in decision-making and optimization tasks, such as automating FEM mesh generation and structural design planning. However, the long training times and challenges in designing reward functions limit its practicality for real-time FEM applications where rapid adaptability is crucial. While each of these models contributes valuable capabilities to FEM-related applications, their limitations, whether in terms of computational requirements, adaptability, or domain-specific challenges, underscore the versatility and practicality of ANNs. Not only these ML models, ANNs also can be identified numerous drawbacks while in the integration with FEM. Hence, the decision on the appropriate ML model should be done considering the purpose and the complexity of the model that is there to be analysed. Further research into the evolving models may complement ANNs, creating hybrid approaches for even more robust FEM solutions.

9. Conclusion and future work

Finite Element (FE) modelling is a highly accurate tool for structural analysis; however, its computational complexity often outweighs its exceptional performance, particularly for real-time applications. To address this limitation, alternative methods such as artificial neural networks (ANNs) have gained significant attention as low-order

surrogates for high-order FE models. ANNs are capable of producing immediate results once trained on extensive datasets that capture a wide range of structural behaviors across varying input parameters or operating conditions. By learning patterns and correlations within these datasets, ANNs can effectively replicate the complex analysis procedures of FE modelling, making them invaluable tools in reducing computational complexity.

The current study has focused on the application of ANNs as surrogates for FE modelling within civil engineering domain, particularly in structural mechanics. Commonly used neural network architectures in this domain include multi-layer perceptron (MLPs), recurrent neural networks (RNNs), convolutional neural networks (CNNs), and graph neural networks (GNNs) are being primarily analyzed in this paper while Physics-informed Neural Networks (PINNs) are also explored and analyzed as a framework that embeds physical laws into the training process, and which can be implemented using these architectures. GNNs demonstrated optimum performance due to their ability to process irregular, spatially varied data from FE meshes. MLPs, with their simplified architecture and strong generalization capabilities, remain popular for structural analysis tasks. However, selecting the optimal neural network type depends on the structural complexity and intended application. Initiating the process with simpler models and progressing to more complex architectures is advisable to mitigate overfitting risks.

While ANNs have shown considerable promise, several main challenges remain unresolved when surrogating FE analysis:

1. Scalability: Developing models that can handle large-scale structures or adapt to diverse structural systems without excessive computational overhead remains a significant hurdle.
2. Generalization: Ensuring trained models generalize effectively to unseen structural configurations and loading scenarios is an ongoing challenge.
3. Physical interpretability: Improving the interpretability of DNN-based predictions, particularly in capturing and preserving the underlying physics of structural behavior, is crucial for broader adoption.

Despite these challenges, emerging opportunities present a promising outlook for NN-based FEM integration. Advanced machine learning techniques, such as PINNs, generative adversarial networks (GANs), Transformers, and diffusion models, offer new possibilities for enhancing model performance. Additionally, hybrid modelling approaches that combine the efficiency of ANNs with the accuracy of FEM could significantly expand their applicability. PINNs, for instance, provide a robust framework for embedding physics-based principles directly into the learning process, although their performance in modelling complex structures requires further refinement. To advance this field, future research should focus on actionable areas, including:

- Developing domain-specific datasets that represent a wide range of structural behaviors.
- Incorporating uncertainty quantification techniques to address variability in material properties, boundary conditions, and loading scenarios.
- Leveraging hardware advancements, such as GPUs, to optimize training efficiency and facilitate real-time simulations.

While ANNs have predominantly been used for forward analyses, less attention has been given to inverse structural analysis. Expanding research in this area could unlock new possibilities for structural optimization and fault detection. Although ANN-based models are typically structure-specific and lack universal applicability, their significantly reduced inference time compared to traditional FEM underscores their potential for real-time applications. With further advancements in machine learning techniques and computational frameworks, ANNs are poised to revolutionize structural analysis, offering faster, scalable, and more efficient solutions for complex engineering challenges.

CRediT authorship contribution statement

S.C. Jayasinghe: Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **M. Mahmoodian:** Writing – review & editing, Supervision, Funding acquisition. **A. Alavi:** Writing – review & editing, Supervision, Investigation. **A. Sidiq:** Writing – review & editing, Validation, Supervision. **F. Shahrivar:** Writing – review & editing. **Z. Sun:** Writing – review & editing. **J. Thangarajah:** Writing – review & editing, Supervision. **S. Setunge:** Writing – review & editing, Supervision.

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Data availability

Data will be made available on request.

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